

Co-expression of the AsCas12a ultra variant, a T7 RNA Polymerase and a cytosine base editor greatly increases transfection and editing rates in *Leishmania* species

Nicole Herrmann May, Annika Schmid, Elisabeth Meiser, Tom Beneke 

Department of Cell and Developmental Biology, Biocentre, University of Würzburg, Am Hubland, 97074 Würzburg, Germany

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

The ability to analyse the function of all genes in a genome has obvious appeal. However, this has been challenging in *Leishmania* due to a repetitive genome architecture, limited DNA repair mechanisms and the absence of RNA interference machinery in most species. While our previous introduction of a cytosine base editor (CBE) tool in *Leishmania* showcased the potential for bypassing these limits (Engstler and Beneke (2023)), challenges remained in achieving high transfection efficiencies, overcoming species-specific editing rates, minimizing effects on parasite growth and eliminating competition between deleterious and non-deleterious mutations. Here, we present an optimized approach to address these limitations. Firstly, we identified a T7 RNAP promoter variant that ensures high editing rates across *Leishmania* species without adversely affecting parasite growth. Secondly, we adjusted the scoring of CBE single-guide RNAs (sgRNAs) to prioritize those ensuring STOP codon generation. Thirdly, we developed a triple-expression construct enabling the integration of CBE sgRNA expression cassettes into a *Leishmania* safe harbor locus via AsCas12a ultra-mediated DNA double-strand breaks. This facilitates the generation of stable CBE sgRNA expression cell lines and increases transfection rates by ~400-fold, resulting in up to one transfectant per 70 transfected cells. Lastly, we show how the co-expression of AsCas12a ultra, T7 RNAP and CBE can be utilized for hybrid CRISPR gene replacement and base editing approaches in the same cell line. Overall, we believe that these improvements will broaden the range of possible gene editing applications in *Leishmania* species and will enable a variety of loss-of-function screens in the future.

eLife assessment

This manuscript describes a method for genetic manipulation of *Leishmania* species which should be sufficiently efficient to enable genome-wide genetic screens. The authors improved numerous aspects of their previously described method, which is based on sequence-specific genome editing to introduce premature stop codons using a CAS9-cytidine deaminase variant. The work is thoroughly described, with **convincing** data, and will be very **important** for *Leishmania* researchers, as well as perhaps suggesting the use of similar approaches in other organisms in which genetic manipulation is challenging.

<https://doi.org/10.7554/eLife.97437.1.sa3>

Introduction

CRISPR/Cas9 gene editing has greatly improved loss-of-function experiments in *Leishmania*, and bar-seq screens have facilitated the functional dissection of large gene cohorts (Baker et al., 2021 [↗](#); Beneke et al., 2019 [↗](#); Beneke and Gluenz, 2019 [↗](#); Beneke and Gluenz, 2020 [↗](#); Beneke et al., 2017 [↗](#); Burge et al., 2020 [↗](#); Damianou et al., 2020 [↗](#)). However, the applications of bar-seq screens are limited, and it remains difficult to target the vast *Leishmania* repertoire of repetitive genetic elements by gene replacement approaches. In addition, *Leishmania* parasites lack crucial components of the non-homologous DNA end joining (NHEJ) pathway (Passos-Silva et al., 2010 [↗](#); Zhang et al., 2022 [↗](#)). As a result, CRISPR-induced DNA double-strand breaks (DSB) lead to unpredictable DNA deletions, increased cell death during DNA repair failures, prolonged repair times and generally low editing rates (Zhang et al., 2017 [↗](#); Zhang and Matlashewski, 2015 [↗](#); Zhang and Matlashewski, 2019 [↗](#); Zhang et al., 2022 [↗](#)). Without other means of selection, such as the selection of drug resistance-associated edits, this complicates further the targeting of dispersed multi-copy genes and applications of pooled CRISPR transfection formats. While RNA interference (RNAi) could offer an alternative (de Paiva et al., 2015 [↗](#); Lye et al., 2022 [↗](#)), its use is confined to species within the *Viannia* subgenus, restricting its applicability in the majority of *Leishmania* species (Lye et al., 2010 [↗](#); Ullu et al., 2004 [↗](#)).

We have recently shown how the use of the hyBE4max cytosine base editor (CBE) (Zhang et al., 2020 [↗](#)) could potentially overcome these limitations. We demonstrated how the conversion of cytosine to thymine, and thereby the introduction of STOP codons, enables the functional disruption of single- and multi-copy genes in *Leishmania* species without requiring DSB, homologous recombination (HR), the additional use of donor DNA, or isolation of clones (Engstler and Beneke, 2023 [↗](#)). Importantly, we presented how this method can be used to identify essential genes in pooled loss-of-function screens via the delivery of plasmid libraries, thereby providing a proof-of-principle experiment for how large-scale pooled CRISPR transfection fitness screens could be carried out in *Leishmania*. However, we also highlighted the need for additional improvements of our method (Engstler and Beneke, 2023 [↗](#)).

For example, since we used a ribosomal promoter derived from *L. donovani* for expression of the CBE and single guide RNA (sgRNA), editing efficiencies varied greatly between species, and for some *Leishmania* parasites, sufficient editing could be only achieved after weeks in culture. Our attempts to use different promoters, such as the T7 RNA Polymerase (RNAP) promoter were unsuccessful, as their employment for CBE sgRNA expression resulted in strong growth defects. Furthermore, the transfection of pooled plasmid libraries led to combinatorial knockout effects, with about 38% of all transfectants harbouring more than one CBE-sgRNA expression plasmid. In

addition, the transfection efficiency of plasmid pools was generally low, yielding only ~1 transfectant out of 30,000 transfected cells. To compensate for combinatorial effects and the relatively low transfection rate, a large number of cells would need to be transfected to enable large-scale pooled CRISPR transfection screens. Lastly, we noticed that non-deleterious mutations within the CBE editing window can become dominant over desired mutations, especially when targeting essential or growth-affecting genes.

Here, we aimed to develop our base editing approach further and deliver needed improvements. We show how we have identified a T7 RNAP promoter variant that enables stable expression of CBE sgRNAs and results in high editing rates within a short period of time without having significant effects on parasite growth. In addition, we present a construct that allows to express not only T7 RNAP and the hyBE4max CBE but also a highly efficient variant of a Cas12a nuclease derived from *Acidaminococcus* sp. (AsCas12a ultra (Zhang et al., 2021 [DOI](#)), formerly AsCpf1). Using AsCas12a ultra-mediated DSBs, we have developed a novel approach to deliver and integrate CBE sgRNA expression cassettes into the *Leishmania* 18S rRNA safe harbor locus (Misslitz et al., 2000 [DOI](#); Sorensen et al., 2003 [DOI](#)). We demonstrate how this has allowed to increase transfection rates by up to 400-fold. In a final optimisation step, we have also improved our CBE sgRNA design and scoring tool www.leishbaseedit.net [DOI](#) to select CBE sgRNAs that have fewer possibilities in generating edits that do not result in a STOP codon. We believe that these optimisation steps now enable us to deliver a range of gene editing applications, including large-scale loss-of-function screens, in *Leishmania* species without limitations due to extreme cases of gene copy numbers, aneuploidy and/or lack of RNAi components.

Results and discussion

T7 RNAP promoter variant-driven CBE sgRNA expression reduces toxicity and improves editing efficiency

In our first version of a CBE toolbox for *Leishmania* we utilized a ribosomal promoter derived from *L. donovani* for CBE protein and sgRNA expression. While achieving high editing rates with this single plasmid system (Figure 1A [DOI](#)) (Engstler and Beneke, 2023 [DOI](#)), efficiencies varied greatly among species, requiring for example over 40 days of incubation to reach significant editing in *L. major* parasites. We previously attempted to minimize these species-specific variations by segregating the CBE expression from the CBE sgRNA cassette in a dual construct system and by employing a T7 RNAP promoter for CBE sgRNA expression (Figure 1B [DOI](#)). However, upon transfection of CBE sgRNA expression constructs into *L. major* T7 RNAP- and CBE-expressing parasites, the doubling time increased from 7 to about 14 hours, highlighting a significant growth defect through T7 RNAP-driven CBE sgRNA expression. On the contrary, we observed higher editing rates in a shorter period of time compared to our approach with the single plasmid ribosomal promoter system, indicating potential advantages (Engstler and Beneke, 2023 [DOI](#)).

Here, our first aim was to eliminate this growth defect observed in the dual construct system while maintaining high editing rates. Assuming that the growth defect resulted from excessive CBE sgRNA expression, we wondered whether modifying the T7 RNAP promoter sequence and altering the transcription initiation site could alleviate the observed toxicity. We therefore introduced a variety of CBE sgRNA expression plasmids into our previously established *L. major* cell line, expressing CBE and T7 RNAP from the β -tubulin locus, as well as tdTomato from the 18S rRNA locus (Engstler and Beneke, 2023 [DOI](#)) (Figure 1B [DOI](#)). To drive CBE sgRNA expression, we thereby utilized two T7 RNAP promoter variants (T7 T-10 and T7 G-10), known for their ability to reduce transcription activity in the closely related parasite *Trypanosoma brucei* (Wirtz et al., 1998 [DOI](#)). We also manipulated the transcription initiation site by altering the number of guanines (either two or one guanines), a factor believed to impact transcription yield (Padmanabhan et al., 2020 [DOI](#)) (Figure 1C [DOI](#)). Additionally, we compared CBE sgRNA expression via T7 RNAP promoters with the

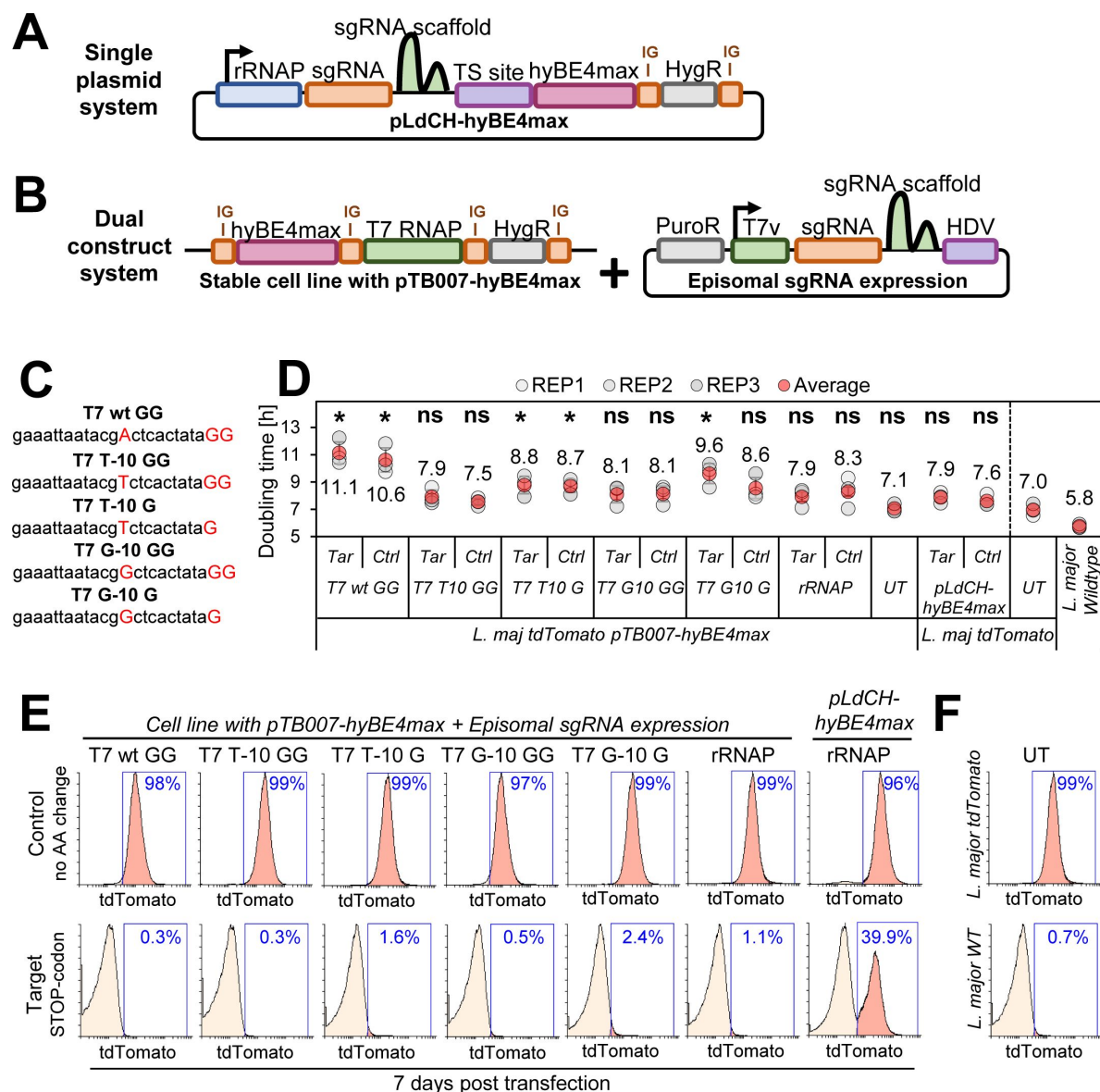


Figure 1.

Optimisation of CBE sgRNA expression in *L. major*.

Schematics of base editing strategies to transfect *Leishmania* parasites either with a (A) single or (B) dual base editing system. (A) Plasmid pLdCH-hyBE4max (Engstler and Beneke, 2023) contains (from left to right) a *L. donovani*-derived ribosomal RNA (rRNA) promoter, single-guide RNA (sgRNA) expression cassette, hepatitis delta virus (HDV) ribozyme containing transsplice sequence (TSS), hyBE4max CBE, *L. donovani*-derived A2 intergenic sequence, and hygromycin resistance marker. (B) Plasmid pTB007-hyBE4max (Engstler and Beneke, 2023) contains separated through intergenic regions (from left to right) a hyBE4max CBE, T7 RNAP and hygromycin resistance marker. This construct is integrated into the β -tubulin locus of *L. major* parasites (LmjF.33:339,096–341,104). The CBE sgRNA expression vector contains a puromycin resistance marker, T7 RNAP promoter, single-guide RNA (sgRNA) expression cassette and HDV ribozyme. (C) Different versions of the T7 RNAP promoter for CBE sgRNA expression have been tested. (D) Doubling times of tdTomato-expressing *L. major* parasites transfected with the single plasmid system and variants of the dual plasmid system for targeting of tdTomato. Error bars show standard deviations of triplicates. Tar: Cells transfected with a CBE sgRNA expression plasmid that facilitates the introduction of a STOP codon within the tdTomato ORF; Ctrl: Cells transfected with a CBE sgRNA expression plasmid that causes a codon mutation without amino acid change. Asterisks indicate Student's t-test: *p > 0.05. (E) FACS plot of parasites shown in (D), analysed 7 days post transfection. (F) FACS plot showing tdTomato-expressing and wildtype *L. major* parasites.

ribosomal promoter derived from *L. donovani*, either in a single or dual construct system setup. For our measurements, we transfected CBE sgRNA expression plasmids with two tdTomato-targeting CBE sgRNAs. While the ‘target’ CBE sgRNA was designed to yield an early STOP codon through C to T conversion, the ‘control’ CBE sgRNA was designed to induce a C to T conversion that results in a neutral substitution (codon change but no amino acid change). Following transfection, we then analysed growth rates (**Figure 1D** [↗](#)) and tdTomato reporter signals (**Figure 2E** [↗](#) and **F** [↗](#)). Our aim was to identify the promoter variant with the highest activity and least toxicity.

As anticipated, when utilizing the ribosomal promoter for CBE sgRNA expression, the dual construct system resulted in a more significant reduction of the tdTomato reporter signal compared to the single plasmid system. The complete knockdown of the tdTomato reporter was also achieved using all T7 RNAP promoter variants with the dual construct system. While there were no detectable differences in these knockdown efficiencies (**Figure 2E** [↗](#) and **F** [↗](#)), there were notable variations in the growth rates (**Figure 1D** [↗](#)). Consistent with our prior observations (Engstler and Beneke, 2023 [↗](#)), cells transfected with constructs containing the unmodified T7 RNAP promoter sequence (T7 wt GG in **Figure 1D** [↗](#)) exhibited an almost two-fold increase in doubling time. On the contrary, employing for CBE sgRNA expression the T7 T10 GG, T7 G10 GG and ribosomal promoter resulted in no significant growth rate increase. The doubling times here ranged from 7.5 to 8.1 hours. While this aligned with measurements obtained for cells transfected with our single plasmid system (**Figure 1D** [↗](#)), the complete knockdown of the tdTomato reporter was much more rapid using any dual construct system (**Figure 1E** [↗](#)). Specifically, the complete depletion of the tdTomato reporter signal was achieved in just seven days (**Figure 1E** [↗](#)) compared to 33 days when using the same CBE sgRNA in *L. major* parasites previously (Engstler and Beneke, 2023 [↗](#)). This demonstrates a major improvement in our method to generate loss-of-function mutants in non-clonal populations.

DSBs introduced by AsCas12a ultra increase integration rates of donor DNA constructs

Having successfully improved editing efficiencies, while at the same time reducing the toxicity of CBE sgRNA expression and targeting, we next attempted to increase transfection rates of the CBE sgRNA expression constructs. Previously, we have shown that ~30,000 *L. mexicana* wildtype cells need to be transfected with our single plasmid system (**Figure 1A** [↗](#)) to obtain one transfectant (Engstler and Beneke, 2023 [↗](#)). Since genomic dropout screens need to be performed at a high representation of about 500 cells per sgRNA construct (Sanjana et al., 2014 [↗](#); Yau and Rana, 2018 [↗](#)), this relatively low transfection rate would make any large-scale screen challenging. In addition, we wanted to enable screening of base editor libraries over long periods and in complex environments, such as *in vivo* experiments, without risking that parasites would lose their CBE sgRNA plasmid, vary their plasmid copy number or transfer plasmids between cells. Therefore, we wanted to integrate CBE sgRNA expression constructs into a safe harbor locus. Strategies to increase transfection rates and stably integrate expression cassettes by homologous recombination have been previously developed in *T. brucei*. Here, the use of an I-SceI mega- (Glover and Horn, 2009 [↗](#)) or zinc finger- (Schumann et al., 2017 [↗](#)) nuclease-induced DSB at a safe harbor locus has been shown to increase the transfection efficiency for the integration of RNAi expression constructs significantly. In combination with improved transfection protocols (Burkard et al., 2007 [↗](#); Glover and Horn, 2009 [↗](#)), this has enabled RNAi loss-of-function screens on a genome-wide scale (Glover et al., 2015 [↗](#); Horn, 2022 [↗](#); Schumann Burkard et al., 2011 [↗](#)). Alternatively, it has been shown in *Leishmania* that CRISPR/Cas9-induced DSBs can increase the rate of correct donor DNA integration, regardless of homology flank length (Beneke et al., 2017 [↗](#)).

To develop a strategy for stable integration of CBE sgRNA expression constructs in *Leishmania* species, we decided to employ a Cas12a nuclease variant from *Acidaminococcus* species. Specifically, we generated an expression construct (pTB107) to co-express in *Leishmania* the AsCas12a ultra variant (Zhang et al., 2021 [↗](#)), a T7 RNAP and the hyBE4max CBE (Engstler and

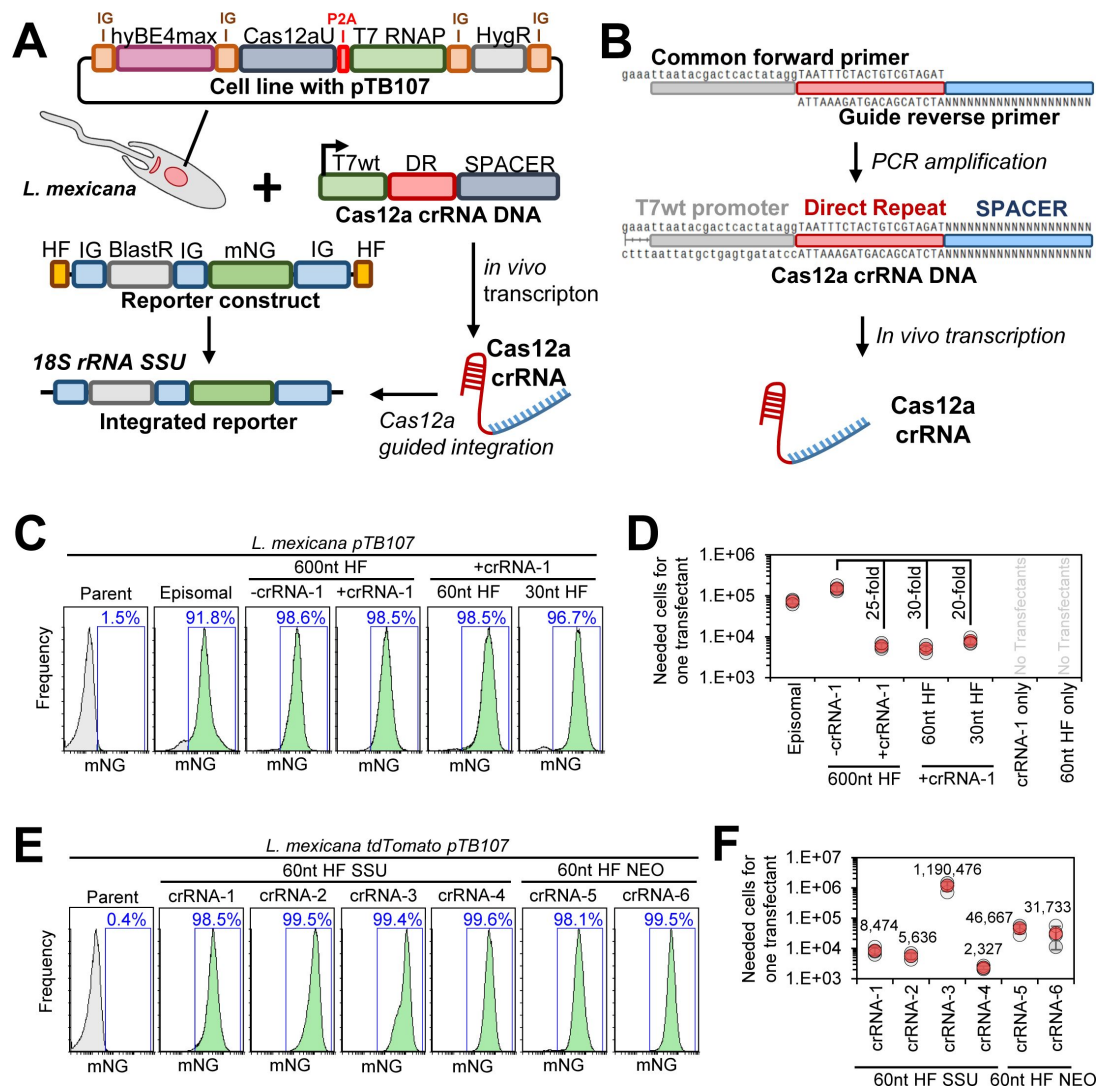


Figure 2.

Developing an AsCas12a-mediated integration system.

(A) Schematic of the Cas12a-mediated integration approach. A stable *L. mexicana* cell line maintains plasmid pTB107 as an episome and is co-transfected with a donor DNA reporter construct and a Cas12a crRNA template DNA. Using a T7 RNAP promoter, the Cas12a crRNA template DNA is *in vivo* transcribed to facilitate a Cas12a-mediated DSB at the 18S rRNA SSU locus. Using homology flanks the reporter construct is efficiently integrated at the site of the DSB. Plasmid pTB107 thereby allows for the expression of hyBE4max CBE, AsCas12a ultra (Zhang et al., 2021), T7 RNAP and hygromycin resistance marker. AsCas12a ultra and T7 RNAP are expressed as a fusion protein but then cleaved through a P2A self-cleaving peptide. The reporter constructs consists of a pPLOT plasmid tagging cassette (Beneke et al., 2017) with additional homology flanks, allowing for the expression of mNeonGreen and a blasticidin resistance marker. (B) Schematic of the Cas12a crRNA template DNA generation. Two overlapping primers are amplified in a PCR reaction. The forward primer is common and contains an unmodified T7 RNAP promoter and an optimised Cas12a direct repeat variant (DR) (DeWeirdt et al., 2021). The guide reverse primer contains the crRNA specific spacer sequence aligning to the targeted locus and the Cas12a DR to enable the overlap with the forward primer. (C and E) FACS plot showing *L. mexicana* pTB107 parasites following the transfection described in (A). Percentages represent the remaining proportion of mNeonGreen-expressing cells. (C) The homology flank (HF) length used for reporter construct integration has been varied and the additional use of Cas12a crRNA was tested. The transfection of a circular plasmid, containing the reporter construct was included as a control (episomal). (E) Six different Cas12a crRNAs have been tested using two different integration loci (as described in the main text). (D and F) Efficiency of transfections shown in (C and E) were measured. The number of cells required for the transfection to obtain one transfectant is shown. Error bars show standard deviations of triplicates.

Beneke, 2023 [↗](#)) (**Figure 2A** [↗](#)). The use of AsCas12a ultra had several advantages compared to other approaches mentioned above. Firstly, compared to the use of I-SceI meganuclease (Glover and Horn, 2009 [↗](#)), it allowed us to select any TTV-PAM containing locus of interest to insert our CBE sgRNA expression construct, simply by just changing the Cas12a crRNA sequence. Secondly, Cas12a editing – just like any CRISPR strategy – is much simpler to design than a zinc-finger nuclease (Schumann et al., 2017 [↗](#)) or TALENs approach. Thirdly, compared to a functional SpCas9 nuclease, there was no risk of interference between the Cas9 base editor and Cas9 nuclease sgRNAs. Finally, since AsCas12a has an intrinsic RNase activity that allows processing of its own crRNA array (Paul and Montoya, 2020 [↗](#)), multiplexing of guide sequences is possible, which could expand the scope of standard CRISPR knockin and knockout approaches in *Leishmania* in future studies. Ultimately, we figured that the co-expression of the AsCas12a ultra nuclease, a T7 RNAP and the CBE would generate a hybrid gene editing toolbox that has many possible applications.

To introduce AsCas12a-mediated DSBs, we decided to adapt the LeishGEdit approach, which was previously developed for delivering Cas9 sgRNAs (Beneke and Gluenz, 2019 [↗](#); Beneke et al., 2017 [↗](#)). Specifically, we designed two overlapping oligos capable of being mutually amplified, generating a Cas12a crRNA template DNA (**Figure 2B** [↗](#)). This resulting template featured an unmodified T7 RNAP promoter sequence, facilitating the *in vivo* transcription of the Cas12a crRNA. To ensure high binding affinity of AsCas12a with its Cas12a crRNA, we used an optimized direct repeat (DR) from DeWeirdt et al. (2021) [↗](#). Subsequently, this enabled AsCas12a ultra-mediated DSBs for donor DNA integration. To test this approach, we utilized a donor DNA construct containing mNeonGreen (mNG) as a reporter and blasticidin as a resistance marker, allowing for a relatively straightforward assessment of integration accuracy and transfection rate (**Figure 2A** [↗](#)). Additionally, we varied the length of homology flanks to investigate whether Cas12a-mediated integration of donor DNA remained independent of homology flank length, as previously observed with Cas9 (Beneke et al., 2017 [↗](#)). Using homology flanks routinely used for integration into the 18S rRNA safe harbor locus of various *Leishmania* species (Misslitz et al., 2000 [↗](#); Sorensen et al., 2003 [↗](#)), we then designed one Cas12a crRNA targeting the 200bp region between both homology regions (**Figure 2 S1A** [↗](#)). To evaluate if Cas12a could enhance the transfection rate, we compared transfections of these reporter constructs with and without the addition of Cas12a crRNA.

Like findings with Cas9, we observed that Cas12a-induced DSBs in *L. mexicana* increased the rate of correct donor DNA integration, irrespective of homology flank length. Integration of reporter constructs with 600 nt homology flanks showed no difference in reporter signal (**Figure 2C** [↗](#)), but the addition of Cas12a crRNA boosted transfection efficiency by approximately 25-fold when assessing blasticidin-resistance rates (**Figure 2D** [↗](#)). When reducing homology flank length to 60 nt, we observed no impact on the reporter signal or transfection rate when Cas12a crRNA was present during transfections. However, in the absence of Cas12a crRNA, no transfectants were obtained, confirming that homologous recombination of the reporter is only independent of homology flank length if a Cas12a-mediated DSB has occurred. Further reducing the homology flank length to 30 nt did not significantly decrease transfection efficiency, but the percentage of mNG positive cells decreased from ~99% in all other samples to 96.7% (**Figure 2C** [↗](#)). Conversely, Sanger sequencing off all integration sites indicated flawless recombination for all constructs (**Figure 2 S1A, B** [↗](#) and **C** [↗](#)). Interestingly, diversity in the reporter signal was also noted when transfecting the mNG construct as an episome (**Figure 2C** [↗](#)), possibly pointing towards plasmid copy number variants between transfected cells. In addition, episomal transfections using the same number of DNA molecules resulted in a significantly lower number of transfectants (**Figure 2D** [↗](#)), concluding the advantages of integrating expression constructs via Cas12a-mediated DSBs.

We then wondered whether employing different Cas12a crRNAs could enhance the transfection efficiency even further. Three additional Cas12a crRNAs within the 18S rRNA SSU locus were selected for testing (**Figure 2 S1D** [↗](#)). In addition, we sought to examine the integration into a “landing-pad”, such as a previously introduced resistance marker. We expected that this would further increase the accuracy and efficiency of the transfection. We therefore decided to test our

system in a tdTomato-expressing *L. mexicana* cell line (Engstler and Beneke, 2023 [↗](#)) and designed two Cas12a crRNAs to integrate mNG-blasticidin donor DNA constructs within their neomycin-resistance marker (**Figure 2 S1D** [↗](#)). Given that 60 nt homology flank length was sufficient for accurate and efficient integration of donor DNA (**Figure 2C** [↗](#) and **D** [↗](#)), we decided to test all six Cas12a crRNAs in combination with donor DNA constructs containing 60nt homology flanks. This meant that homology flanks could be easily adapted through long-primer PCR without the need for bacterial cloning.

Across all six Cas12a crRNAs tested, no significant changes in the proportion of mNG-expressing cells were observed (**Figure 2E** [↗](#)). However, there was a substantial variation in the rate of blasticidin-resistant cells, indicative of Cas12a crRNA efficiency (**Figure 2F** [↗](#)). For Cas12a crRNA-1, we confirmed our previous results, yielding one transfectant per ~8,500 transfected cells. However, other Cas12a crRNAs, particularly Cas12a crRNA-4, performed significantly better, resulting in one transfectant out of approximately 2,300 transfected cells. Interestingly, despite being only 44bp upstream of Cas12a crRNA-4, Cas12a crRNA-3 exhibited a 500-fold lower integration efficiency. Surprisingly, both neomycin-targeting Cas12a crRNAs were less efficient than the majority of 18S rRNA SSU locus-targeting guides (**Figure 2F** [↗](#)). Given the exceptional efficiency of Cas12a crRNA-4, we verified the integration of donor DNA at its site through Sanger sequencing, confirming flawless integration (**Figure 2 S1D, E** [↗](#), and **F** [↗](#)).

Integration of CBE sgRNA expression cassettes via AsCas12a ultra-introduced DSBs increase editing rates

After successfully validating our here developed AsCas12a ultra-mediated integration system, we then decided to combine our Cas12a knock-in system with our optimized CBE sgRNA expression cassette (**Figure 3A** [↗](#)). For this purpose, we developed two CBE sgRNA expression plasmids, namely pTB104 and pTB105, each incorporating a puromycin-resistance marker and the most effective T7 RNAP promoter variant (T7 T10 GG) for CBE sgRNA expression (as identified in assays above; **Figure 1C, D** [↗](#) and **E** [↗](#)). While pTB104 featured 350 nt homology flanks adjacent to the neomycin-targeting Cas12a crRNA-6, pTB105 included 350 nt homology flanks adjacent to our best-performing Cas12a crRNA-4, targeting the 18S rRNA SSU locus. This design allowed for the integration of both CBE sgRNA expression cassettes with and without the addition of Cas12a crRNAs. To assess if such a system could effectively generate a functional null mutant through cytosine-to-thymine conversion, we chose to target tdTomato again and tested a tdTomato-targeting CBE sgRNA in *L. donovani*, *L. mexicana*, and *L. major*. Parasites expressed tdTomato, along with the CBE, Cas12a ultra, and T7 RNAP (**Figure 3A** [↗](#)). Additionally, we investigated whether the efficiency of the tdTomato knockdown would be influenced by how the CBE sgRNA expression cassette was integrated. We therefore compared Cas12a crRNA-4 and 6 in combination with 60 and 350 nt homology flanks.

Strikingly, we observed a complete depletion of the tdTomato signal just 7 days post transfection in all tested species when integrating CBE sgRNA expression cassettes into the 18S rRNA SSU locus. This occurred irrespective of homology flank length. However, integrating cassettes into the neomycin resistance marker resulted in only partial tdTomato knockdowns in *L. major* and *L. donovani*, while a complete knockdown could again be achieved in *L. mexicana*. Although we have not further investigated the cause for this, we assume that either the low efficiency of Cas12a crRNA-6 or the number of tdTomato-neomycin-expression cassettes caused this discrepancy. Overall, this highlights that integrating CBE sgRNA expression cassettes into the 18S rRNA SSU locus using Cas12a crRNA-4 in combination with plasmid pTB105 would yield the highest CBE sgRNA editing efficiency. We therefore tested next the transfection rate of CBE sgRNA expression cassettes using Cas12a crRNA-4 for integration in *L. mexicana*, *L. major* and *L. donovani*. We reached high transfection efficiencies, yielding up to one transfectant out of ~470 transfected *Leishmania* parasites, with *L. donovani* being the most and *L. major* being the least efficient (**Figure 3C** [↗](#), left panel).

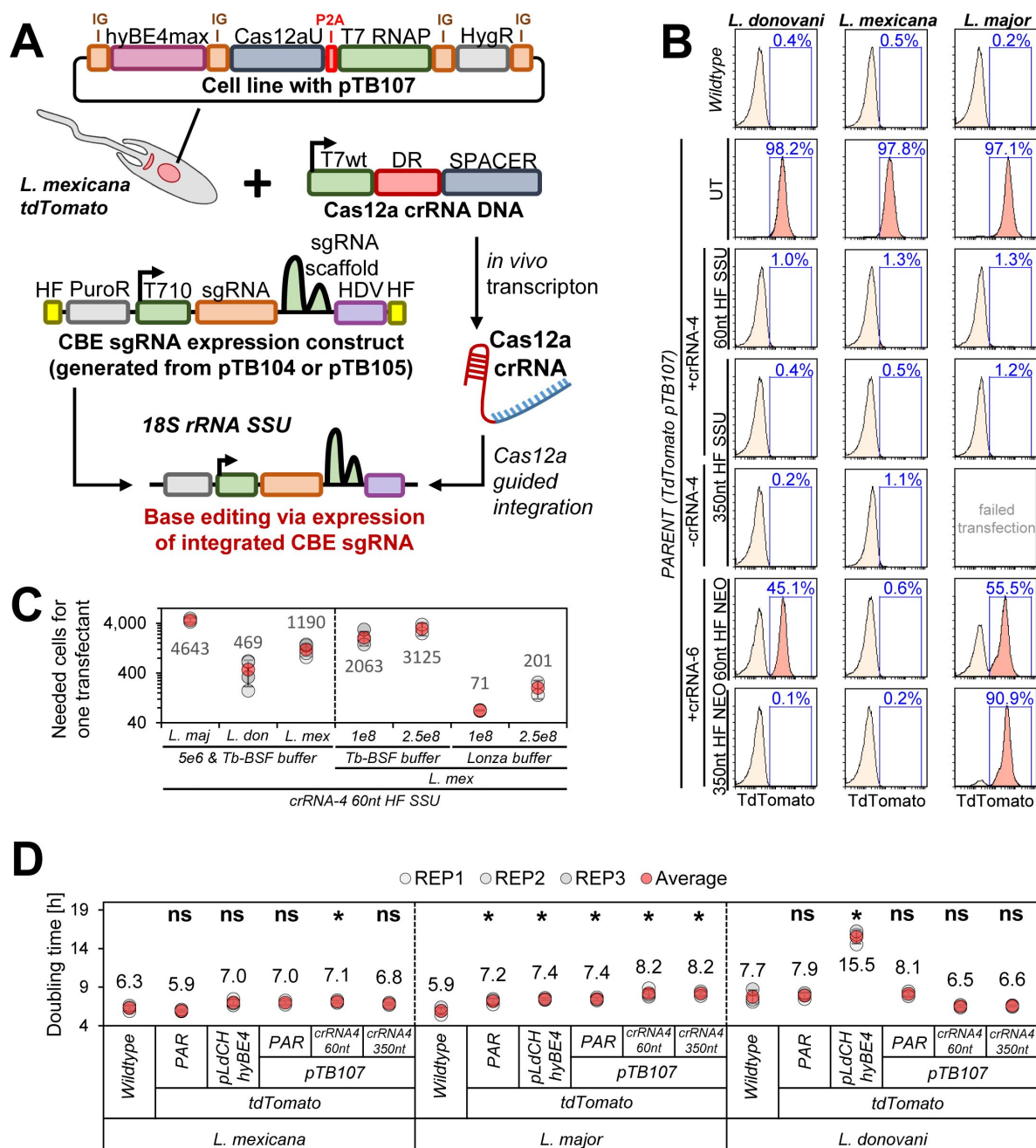


Figure 3.

Integration of CBE sgRNA expression cassettes via AsCas12a ultra.

(A) Schematic of Cas12a-mediated integration of the CBE sgRNA expression construct. The pTB107 stable cell line is co-transfected with two constructs: (1) a Cas12a crRNA template DNA and (2) a CBE sgRNA expression construct. The Cas12a crRNA template DNA is *in vivo* transcribed and consists of an unmodified T7 RNAP promoter (T7wt, light green), a Cas12a direct repeat (DR, red) and a 20 nt Cas12a guide target sequence (SPACER, blue). The CBE sgRNA expression construct is integrated into the 18S rRNA SSU locus, following the Cas12a-mediated DSB. This donor construct contains two homology flanks (HF, yellow), a puromycin-resistance marker (grey), a T7 T-10 GG promoter (green), a guide target sequence (orange), a Cas9 scaffold (dark green) and an HDV (purple). (B) FACS plots show pTB107 parasites that express tdTomato and have been transfected with pTB104 and pTB105 CBE sgRNA expression cassettes, containing a tdTomato-targeting guide. Percentages represent the remaining proportion of tdTomato-expressing cells. (C) Following dilutions after transfection, the number of puromycin-resistant transfectants obtained per transfectant cell was calculated. Error bars show standard deviations of triplicates. (D) Doubling times for transfectants shown in (B). PAR: parental cell line, UT: Un-transfected control.

We then wondered whether we could increase the transfection efficiency even further by using the Lonza Nucleofector technology. Previous studies in *T. brucei* demonstrated that the improved transfection rate achieved through I-SceI meganuclease-mediated cleavage could be further increased when combined with the Lonza Nucleofector technology, resulting in up to one transfectant per 100 transfected cells (Glover and Horn, 2009). Considering this, we explored whether this approach would be as effective in *Leishmania* parasites. We therefore conducted a site-by-site comparison of the transfection efficiency between the Lonza Nucleofector Basic Parasite kit and the *Trypanosoma*-optimized buffer (Tb-BSF buffer, see method section for further details) (Schumann Burkard et al., 2011), both in combination with an Amaxa Nucleofector 2b electroporator. To increase the number of parasites per transfection even further, we compared our standard transfection protocol using 5×10^6 cells per transfection with up-scaled transfection formats involving 1×10^8 cells and 2.5×10^8 cells per electroporation. Testing these conditions in *L. mexicana*, we confirmed that the Lonza Nucleofector technology could indeed boost transfection efficiency by an additional 20-fold (Figure 3C, right panel). When transfecting 1×10^8 cells per transfection with 10 μ g of PacI-linearized pTB105 plasmid DNA, we achieved up to one transfectant out of 70 transfected cells (Figure 3C, right panel). This represents a more than 400-fold increase in transfection efficiency compared to our previous rates (Engstler and Beneke, 2023). Moreover, it demonstrates that a CBE sgRNA library with 40,000 constructs could be transfected at a representation rate of 500-fold using just 20×10^8 cells across 20 transfection cuvettes. Ensuring a 500-fold library representation represents a substantial improvement compared to previous RNAi screens in *T. brucei*, where screens typically achieved about 5-9-fold library representation (Glover et al., 2015; Horn, 2022; Morris et al., 2002; Schumann Burkard et al., 2011). This high representation rate is considered crucial for large-scale dropout screens, as it can affect hit identification by minimizing variations between replicates. However, it's worth noting that drug resistance screens can also be conducted at much lower coverage (Sanjana et al., 2014; Yau and Rana, 2018).

While the significantly enhanced transfection rates are a central aspect of our improved base editing method presented here, it is crucial to emphasize that these improvements were achieved without impacting parasite growth. Through an assessment of doubling times in transfected and parental cell lines, we confirmed that the expression of CBE, T7 RNAP, AsCas12a ultra, and the CBE sgRNA had minimal or no impact on parasite growth across all three species (Figure 3D). This marks a substantial improvement, considering that the expression of CBE and/or CBE sgRNA in our first toolbox version, using the pLdCH-hyBEmax plasmid, could lead to significant growth defects in some *Leishmania* species (Engstler and Beneke, 2023).

Having established this optimised system, we next decided to target PF16, a gene encoding a central pair protein of the axoneme that has been demonstrated to be essential for *Leishmania* flagellar motility (Beneke et al., 2017; Engstler and Beneke, 2023; Martel et al., 2017). For our test, we chose to utilize a previously employed PF16-targeting CBE sgRNA, namely PF16-3. This specific CBE sgRNA induces paralysis in *Leishmania* parasites by introducing a thymidine homopolymer ("TTTTT") within the PF16 coding sequence. While our usual design for CBE sgRNAs aims to introduce STOP codons through cytosine-to-thymine conversion, we selected PF16-3 for this test because its editing activities are known to vary across species (Engstler and Beneke, 2023). In earlier attempts, we had to express CBE sgRNA PF16-3 for 42 days to achieve sufficient editing in *L. major* when using our pLdCH-hyBE4max single vector (Figure 1A) (Engstler and Beneke, 2023). Meanwhile, in *L. donovani*, high editing rates were already achieved 14 days post transfection. Now, we expected that editing rates had improved uniformly across all species.

As anticipated, we observed a significant increase in editing activity in *L. major* parasites. Just 14 days post transfection, we found that the homo-polymer had been fully introduced when using our T7 RNAP promoter-based dual vector system, while no editing could be observed using the

ribosomal promoter single vector system at this time point (**Figure 4B** [↗](#), *L. major* panel). This was also reflected in the analysis of mutant swimming speed, revealing clear motility defects in respective transfectants (**Figure 4A** [↗](#), *L. major* panel).

We next proceeded to test our optimized system also in *L. mexicana* and *L. donovani* parasites, once again employing Cas12a for integration of the CBE sgRNA expression construct. However, here we chose to target PF16 in two cell lines: (i) one that possessed the pTB107 construct and a tdTomato reporter, and (ii) one that possessed the pTB107 construct only (WT pTB107). Additionally, we took this opportunity to compare motility and CBE editing rates in cells that had been transfected with Cas12a crRNAs that target the neomycin-resistance marker (Cas12a crRNA-6) and the 18S rRNA SSU locus (Cas12a crRNA-4).

Our results clearly showed that just 6 days post transfection, the PF16 target site was fully edited, and the overwhelming majority of all cells were paralyzed in all tested scenarios (**Figure 4A** [↗](#) and **B** [↗](#)). However, surprisingly editing slowly reversed in *L. mexicana* cells harboring the pTB107 construct only (**Figure 4A** [↗](#) and **B** [↗](#)). Although we decided not to further investigate the reason for this unexpected result, we assumed that it might be caused by a small proportion of cells where the CBE sgRNA expression cassette was incorrectly integrated. Since the PF16 mutation in *L. mexicana* is known to be associated with a mild growth defect ([Beneke et al., 2019](#) [↗](#)), PF16-deficient mutants would eventually be outcompeted over time, leading to the reversal of editing rates. But this phenomenon was not observed in any other tested cell line, when comparing editing and motility rates 6 and 16 days post transfection. Both tested *L. donovani* cell lines and the tested *L. mexicana* cell line that possessed the pTB107 construct and the tdTomato expression cassette (**Figure 4A** [↗](#) and **B** [↗](#)) remained paralyzed over the entire observation time. To confirm that these cells would have no potential ever to reverse their phenotype, we decided to Illumina sequence paralyzed *L. mexicana* PF16 mutants 16 days post transfection. Analyzing their sequencing data, we did not find a single read of an unedited PF16 CDS sequence (with approximately 35× genome coverage), confirming that indeed all cells were completely mutated (**Figure 4** [↗](#) **S1A** [↗](#)).

Cas12a-mediated DSB ensures the integration of one CBE sgRNA per *L. mexicana* transfectant

We next wanted to verify how exactly the CBE sgRNA expression cassette had been integrated via Cas12a-mediated DSB in *L. mexicana* cells that possessed the pTB107 construct and the tdTomato reporter. Since the tdTomato expression cassette is just ~400 nt upstream of the Cas12a crRNA-4 target sequence, we wondered whether it had come to any obvious interference between the tdTomato and CBE sgRNA expression cassettes. Given the highly repetitive nature of the 18S rRNA SSU locus, we employed Oxford Nanopore Technology (ONT) sequencing to determine the exact integration pattern of both cassettes. For the analysis of sequencing reads, we mapped them against two customized genomes. In the first genome, we assumed that the tdTomato expression cassette was integrated separately from the CBE sgRNA construct, implying that each cassette had integrated on the opposite allele (**Figure 4** [↗](#) **S1B** [↗](#)). In the second genome, we assumed that both cassettes were integrated on the same allele and could be detected on a single ONT read (**Figure 4** [↗](#) **S1C** [↗](#)).

Interestingly, we could only map unique ONT reads to the first scenario, where cassettes were assumed to be integrated on separate alleles and present in only one copy in the genome. These ONT reads covered the entire tdTomato or CBE sgRNA expression cassette, as well as sequences adjacent to homology flanks used for integration (**Figure 4** [↗](#) **S1B** [↗](#) and **C** [↗](#)). To validate our findings, we revisited our Illumina sequencing data and analyzed the coverage of relevant genetic features. Specifically, we normalized all reads to the total number of reads per chromosome and compared the normalized coverage of reads mapped to the CDS of PF16, the CBE sgRNA cassette,

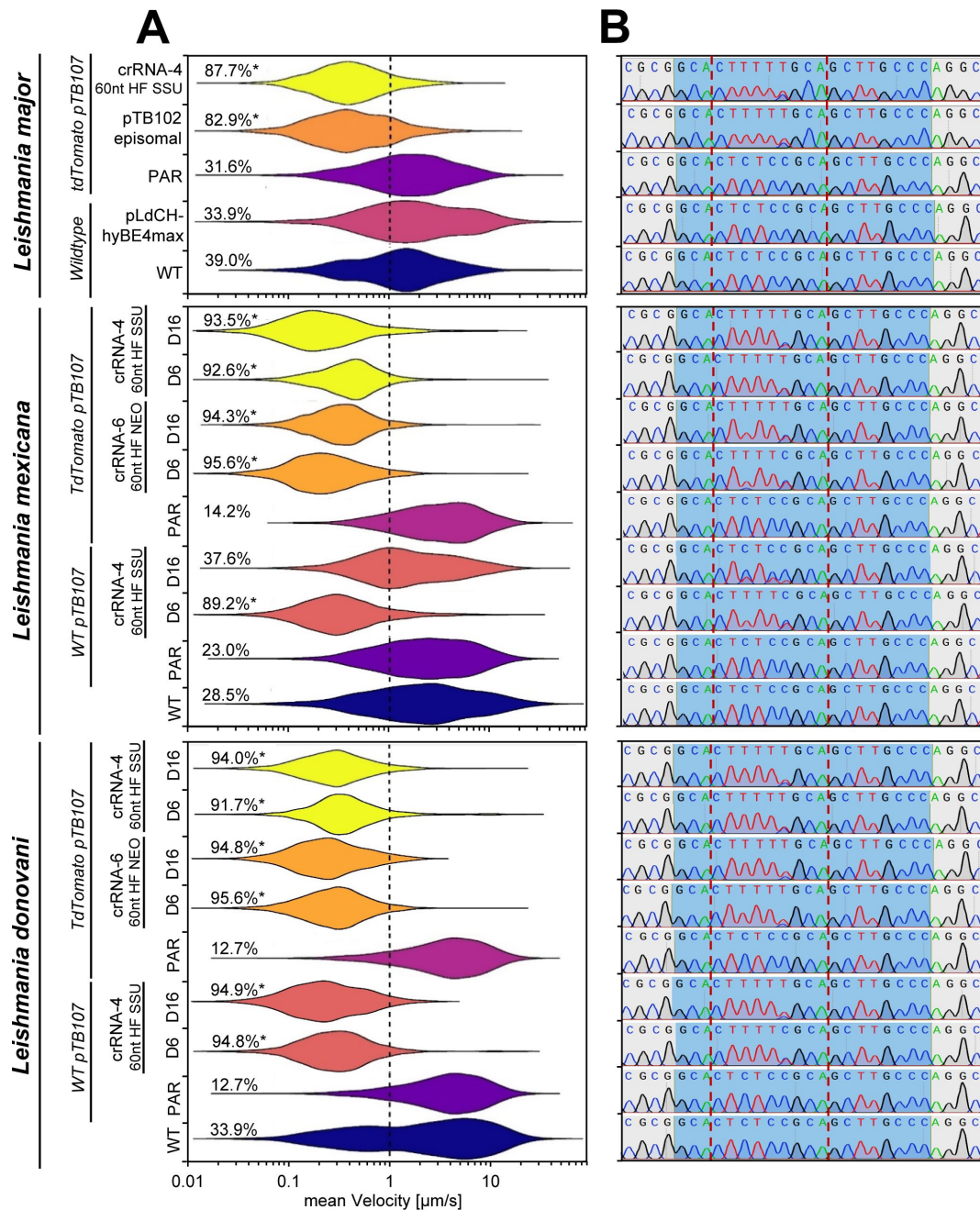


Figure 4.

Targeting PF16 via Cas12a-delivered CBE sgRNA expression cassettes.

L. major, *L. mexicana*, *L. donovani* wildtype and tdTomato/pTB107 cell lines were transfected with a range of CBE sgRNA expression constructs in order to functionally mutate PF16. Different strategies for construct delivery and sgRNA expression were tested, including the integration of constructs into the neomycin-resistance marker (using Cas12a sgRNA-6) and the 18S rRNA SSU locus (Cas12a sgRNA-4), each using 60 nt homology flanks. For comparison to our previous system (Engstler and Beneke, 2023), we also expressed CBE sgRNAs from an episome and transfected pLdCH-hyBE4max into *L. major*. (A) Violin plot of pooled replicates from motility tracked non-clonal populations. The mean velocity of tracked cells was plotted 14 days post transfection for *L. major*, and 6 and 16 days post transfection for *L. mexicana* and *L. donovani*. The total percentage of tracked cells showing a velocity of less than 1 μm/s is highlighted. Each population was analysed using a Cramér-von Mises Test to detect any shift in the population distribution towards lower speed. Percentages are marked with an asterisks when that shift was significant (*p > 0.05). (B) Corresponding Sanger sequencing trace plots. Blue shading: 20 nt guide target sequence. Red dotted lines: hyBE4max editing window.

the CDS of tdTomato, and the CDS of the neomycin-resistance marker. While we found equal read coverage of PF16-mapped reads and the chromosome harboring PF16, read coverage for the CBE sgRNA cassette, the CDS of tdTomato, and the CDS of the neomycin-resistance marker were approximately half of the read coverage of their harboring chromosome 27 ([Figure 4 S1D](#)). This confirmed our hypothesis that indeed the CBE sgRNA expression cassette had only been integrated on one allele, and most likely, this meant that only one CBE sgRNA cassette copy was present per *L. mexicana* cell.

This represents a major improvement since, in our previously used single vector system, about 38% of all *L. mexicana* transfectants possessed more than one CBE-sgRNA expression plasmid, potentially causing combinatorial knockout effects. To verify that in our new dual system each cell indeed harboured only one CBE sgRNA, we decided to test if isolated clones from a transfected library would possess one or multiple CBE sgRNA sequences. In addition to testing this in *L. mexicana*, we also wanted to assess this for *L. donovani* and *L. major*, as we had not sequenced respective mutants via ONT and Illumina sequencing.

Using again Cas12a crRNA-4 for the integration of a small-scale pTB105 CBE sgRNA expression cassette library (containing 13 different CBE sgRNA sequences), we isolated 10 clones from each species, PCR amplified the integrated CBE sgRNA locus, and sequenced it by Sanger sequencing. As expected, we found only unique sequencing reads for all tested *L. mexicana* clones, confirming the integration of only one guide per transfectant ([Figure 4 S2](#)). However, for *L. donovani* and *L. major*, 2 out of 10 sequencing reads showed mixed guide sequences ([Figure 4 S2](#)), suggesting that multiple CBE sgRNAs can be integrated per cell, though at a slightly lower rate of about 20% compared to the previous 38% observed with plasmid transfections. Furthermore, the integration of multiple sgRNAs per cell remained random. While testing other Cas12a crRNAs in the future may identify a more unique integration locus for *L. donovani* and *L. major* parasites, any compensatory effects through the integration of multiple CBE sgRNAs in these species will likely be minimal and presumably not detectable in large-scale screens with a high library representation rate. Therefore, we believe that this method is suitable not only for screens in *L. mexicana* but also in *L. donovani* and *L. major*.

Improved CBE sgRNA design to prioritize edits resulting only in STOP codons

The results above demonstrate that Cas12a crRNAs can be used to efficiently integrate CBE sgRNA expression cassettes into *L. mexicana*, *L. major* and *L. donovani*. This has no significant impact on parasite growth and enables high editing rates in all these tested species. However, as we noted in our previous study ([Engstler and Beneke, 2023](#)), high editing rates do not necessarily lead to a strong knockdown of the targeted protein. Cells with a deleterious mutation can be simply outcompeted by cells with a non-deleterious mutation. This issue is particularly relevant when not all available cytosine nucleotides within the CBE sgRNA editing window result in a STOP codon after thymine conversion. To address this, we have therefore improved the scoring of previously designed CBE sgRNAs by implementing a C-score that determines the ratio of available cytosines to resulting STOP codons ([Figure 4 S3](#)). This new scoring system has been uploaded to www.leishbaseedit.net. We also considered the position at which the STOP codon would occur and designed a scoring matrix that rates CBE sgRNAs higher when the STOP codon is introduced within positions 4-8 and lower when introduced within positions 11-12 ([Figure 4 S3](#)). While our ratio scoring aims to increase the chance of successfully generating a functional mutation through STOP codon insertion, it is important to note that the introduction of a STOP codon is not the only efficient way to generate a functional mutant. As shown above, the introduction of a thymidine homo-polymer ("TTTTT") within the PF16 CDS produces the same motility phenotype observed when using gene replacement strategies for knockout generation ([Beneke et al., 2017](#)). Ultimately, this expands the possibilities for upcoming mutational screens in *Leishmania*.

Co-expression of Cas12a and CBE enables a wide range of genetic manipulations

While these modifications to our sgRNA scoring tool represent another improvement in our method, the design of our triple expression construct (AsCas12a ultra, T7 RNAP, and Cas9 CBE) also offers opportunities for other advancements. For example, it enables hybrid approaches where classical CRISPR gene replacement and/or tagging methods can be combined with cytosine base editing experiments in one cell line. This is particularly interesting for targeting multiple genes simultaneously or dispersed multi-copy genes, and can also include the co-targeting of non-coding and coding genetic elements. Furthermore, since Cas12a has an intrinsic RNA cleavage activity, multiplexing of sgRNAs is possible and this expands the range of potential applications further.

To explore the applicability of AsCas12a ultra for gene replacement strategies, such as the LeishGEdit approach (Beneke and Gluenz, 2019 [DOI](#); Beneke and Gluenz, 2020 [DOI](#); Beneke et al., 2017 [DOI](#)), we devised two Cas12a crRNAs targeting the 5' and 3' UTR of the PF16 ORF (**Figure 5A** [DOI](#)). Additionally, we designed 30nt homology flanks adjacent to these crRNAs. We then PCR-amplified pTBlast and pTPuro cassettes from the LeishGEdit toolbox using primers containing these homology flanks and co-transfected them with Cas12a crRNA template DNA into *L. mexicana* parasites. These cells contained the pTB107 construct for co-expression of AsCas12a ultra, T7 RNAP, and CBE. Like the Cas9 LeishGEdit approach, transfected Cas12a crRNAs contained a T7 RNAP promoter for efficient *in vivo* transcription (**Figure 2B** [DOI](#)). Following transfection, we then confirmed that the ORF was successfully replaced by drug resistance markers on both alleles and amplified the PF16 locus with two ORF spanning primers (**Figure 5A** [DOI](#) and **B** [DOI](#)). In addition, we performed Sanger sequencing over integration flanks to verify the integration of donor DNA fragments. This analysis confirmed the accurate integration of both donor DNA constructs (**Figure 5C** [DOI](#) and **D** [DOI](#)). Lastly, we performed again our motility analysis on these mutants, revealing the expected PF16 motility phenotype (**Figure 5E** [DOI](#)). Overall, this demonstrates the versatility of our AsCas12a ultra, T7 RNAP, and CBE co-expression vector for a wide range of CRISPR applications in *Leishmania*.

Conclusion

The *Leishmania* research community faces significant challenges in targeting all genes within the species. The absence of a functional NHEJ pathway complicates the process of obtaining null mutants, typically demanding the additional use of donor DNA, selection of edits associated with drug resistance, or the time-consuming isolation of clones. These complexities also apply to loss-of-function screens. Classical pooled CRISPR transfection screening formats are unlikely to be executed successfully, unless additional selective measures are used. While, bar-seq screens (Baker et al., 2021 [DOI](#); Beneke et al., 2019 [DOI](#); Beneke and Gluenz, 2020 [DOI](#)) and RNAi methods (de Paiva et al., 2015 [DOI](#); Lye et al., 2022 [DOI](#)) offer potential solutions, they come with clear limitations. Most notably, the use of RNAi is restricted to *Leishmania* species within the *Viannia* subgenus, and bar-seq screens pose logistical challenges due to the necessity of generating thousands of individually created mutants. Furthermore, CRISPR gene replacement approaches, such as bar-seq methods, require multiple rounds of transfection when targeting dispersed multi-copy genes and are not suited to mutate interrupted tandemly arrayed coding and non-coding genes, which are abundantly present in the *Leishmania* genome.

Therefore, we recently introduced cytosine base editing as an alternative method for gene editing and large-scale screening in *Leishmania*, aiming to overcome these limitations. However, variations in editing efficiency, substantial reduction in parasite growth, competition between

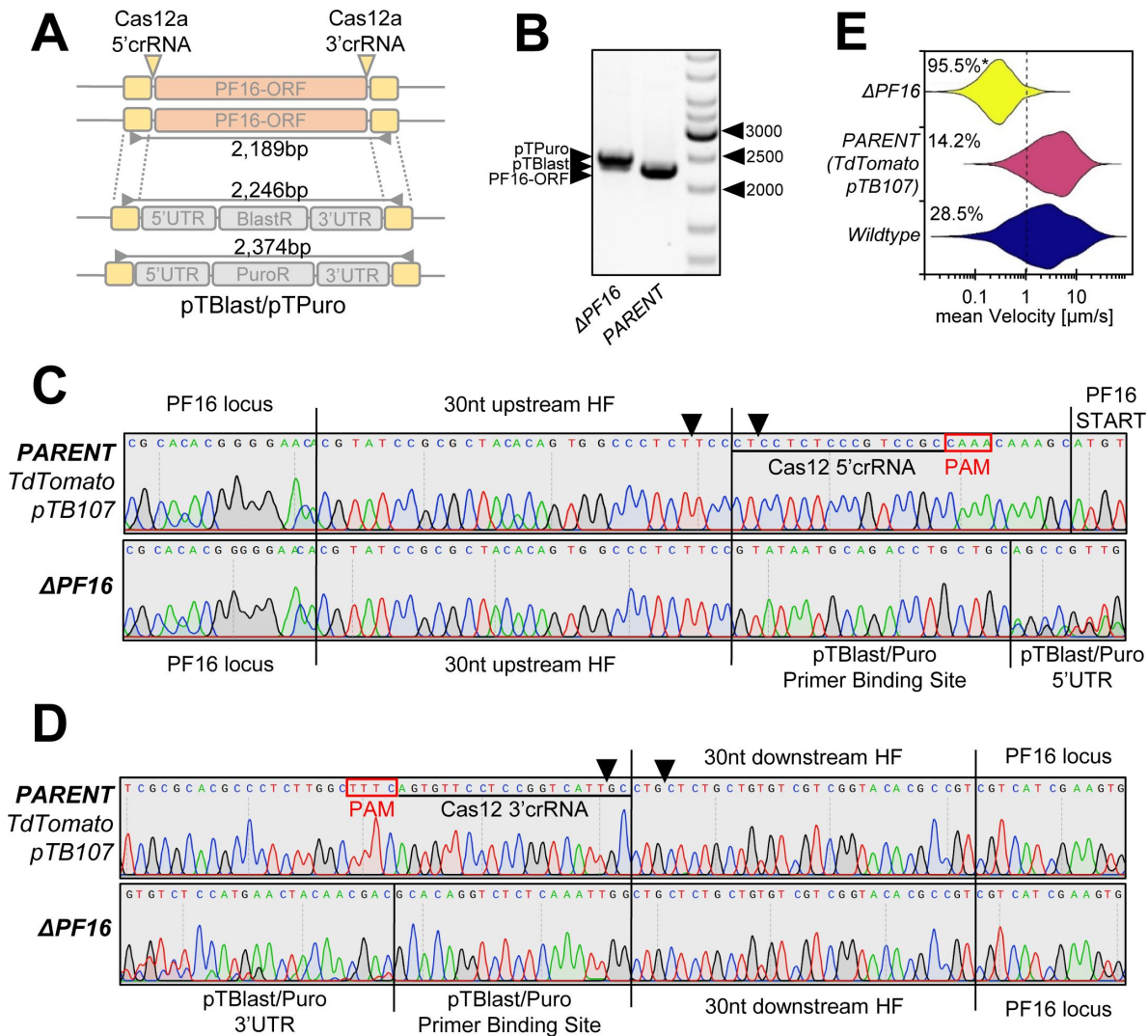


Figure 5.

PF16 gene replacement by using CRISPR/AsCas12a ultra.

(A) Map of the PF16 locus before and after targeting with Cas12a 5' and 3'crRNA positions indicated. The PF16-ORF is replaced with a pTBlast and pTPuro cassette. Primers used for verification of the ORF replacement are highlighted with PCR amplicon length. (B) Visualization of PCR amplicons, showing expected products for pTPuro and pTBlast in the ΔPF16 lane and the amplicon of the PF16-ORF in the PARENT lane (*L. mexicana* tdTomato/pTB107 cell line). (C and D) Sanger sequencing trace plots of amplicons in (B), showing sequencing of the upstream region in (C) and downstream region in (D). Black arrows indicate Cas12a DSBs. (E) Violin plot of pooled replicates from motility tracked non-clonal populations, including the ΔPF16 and PARENT cell line. The total percentage of tracked cells showing a velocity of less than 1 μm/s is highlighted. Each population was analysed using a Cramér-von Mises Test to detect any shift in the population distribution towards lower speed. Percentages are marked with an asterisks when that shift was significant (*p > 0.05).

deleterious and non-deleterious mutations, and low transfection rates prompted further refinements of our method. Here we present major improvements that collectively enhance our CBE method for *Leishmania* parasites.

In our final setup, we effectively employ a dual construct system, where a stable cell line is transfected with a CBE sgRNA expression construct. The stable cell line expresses the hyBE4max CBE, AsCas12a ultra, and T7 RNAP, with CBE sgRNA expression driven by a T7 RNAP promoter variant (T7 T-10 GG). Alongside our updated CBE sgRNA design tool, which now prioritizes sgRNAs with a low cytosine to resulting STOP codon ratio, this system achieves high editing rates without affecting parasite growth. Compared to our initial CBE single-vector method, this dual-construct system has the disadvantage that it requires the generation of a stable cell line first. However, it offers the clear advantage that CBE sgRNA expression constructs can be efficiently integrated using AsCas12a-mediated DSBs. Through testing a series of Cas12a crRNAs, we identified a specific Cas12a crRNA capable of efficiently integrating CBE sgRNA expression cassettes into the 18S rRNA SSU safe harbor locus. CBE sgRNA expression cassettes can thereby be generated by amplification or digestion of the plasmid pTB105. In combination with the Lonza Nucleofector Technology, this approach increases transfection rates by ~400-fold, yielding up to 1 transfectant per 70 transfected cells. Additionally, the co-expression of Cas12a and CBE provides strategic advantages, enabling the use of CRISPR gene replacement and base editing approaches in the same cell line. Moreover, Cas12a can be easily employed for protein tagging and multiplexing of sgRNAs, offering in the future a wide range of possible editing experiments in combination with base editing. Overall, we believe that our improved toolbox sets the stage for various gene editing applications in *Leishmania*, including genome-wide cytosine base editor screens, and we hope that these will make significant advancements in the field.

Material and methods

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (<i>Leishmania mexicana</i>)	<i>L. mexicana</i> wildtype	Eva Gluenz laboratory	WHO strain MNYC/BZ/62/M379	Used TriTrypDB (release 59, (Aslett et al., 2010)) reference annotation: <i>L. mexicana</i> MHOMGT2001U1103
Cell line (<i>Leishmania major</i>)	<i>L. major</i> wildtype	Eva Gluenz laboratory	strain Friedlin	Used TriTrypDB (release 59, (Aslett et al., 2010)) reference annotation: <i>L. major</i> Friedlin
Cell line (<i>Leishmania donovani</i>)	<i>L. donovani</i> wildtype	Joachim Clos laboratory (Decuypere et al., 2005)	strain BPK190	Used TriTrypDB (release 59, (Aslett et al., 2010)) reference annotation: <i>L. donovani</i> BPK282A1
Recombinant DNA reagent	pTB007-hyBE4max	(Engstler and Beneke, 2023)		Contains CBE hyBE4max and T7 RNAP
Recombinant DNA reagent	pLdCH-hyBE4max	(Engstler and Beneke, 2023)		Contains CBE hyBE4max and CBE sgRNA
Recombinant DNA reagent	pLdCH-hyBE4max-LmajDBD	(Engstler and Beneke, 2023)		Contains Leishmania-optimised CBE hyBE4max and CBE sgRNA
Recombinant DNA reagent	pTB107	This study		Contains CBE hyBE4max, AsCas12a ultra and T7 RNAP
Recombinant DNA reagent	pTB106	This study		Contains Leishmania-optimised CBE hyBE4max, AsCas12a ultra and T7 RNAP
Recombinant DNA reagent	pTB102	This study		Contains T7 RNAP promoter-driven CBE sgRNA expression cassette without homology flanks
Recombinant DNA reagent	pTB104	This study		Contains T7 RNAP promoter-driven CBE sgRNA expression cassette with homology flanks for neomycin-resistance gene

Recombinant DNA reagent	pTB105	This study		Contains T7 RNAP promoter-driven CBE sgRNA expression cassette with homology flanks for 18S rRNA SSU locus
Software	TriTrypDB (release 59)	(Aslett et al., 2010)	https://tritrypdb.org/tritrypdb/app	
Software	LeishBASE edit	This study	http://www.leishbaseedit.net/	See description under “Automated CBE guide design using LeishBASEedit”
Gene (<i>L. donovani</i> BPK282A1, <i>L. major</i> Friedlin, <i>L. mexicana</i> MHOMGT2 001U1103)	PF16	TriTrypDB (release 59) (Aslett et al., 2010)	LdBPK_201450.1, LmjF.20.1400, LmxM.20.1400	

Cell culture

As described in Engstler and Beneke (2023) [\[1\]](#), promastigote-form *L. mexicana* (WHO strain MNYC/BZ/62/M379), *L. major* Friedlin and *L. donovani* (strain BPK190, (Decuypere et al., 2005 [\[2\]](#))) were grown at 28°C in M199 medium (Life Technologies) supplemented with 2.2 g/L NaHCO₃, 0.0025% haemin, 0.1 mM adenine hemisulfate, 1.2 µg/mL 6-biopterin, 40 mM 4-(2-Hydroxyethyl)piperazine-1-ethanesulfonic acid (HEPES) pH 7.4 and 20% FCS. Media were supplemented with the relevant selection drugs: 40 µg/ml Hygromycin B, 40 µg/ml Puromycin Dihydrochloride and 40 µg/ml G-418 Disulfate. The identity of each *Leishmania* species and absence of mycoplasma contamination was confirmed previously (Engstler and Beneke, 2023 [\[1\]](#)). Doubling times were determined as described in Engstler and Beneke (2023) [\[1\]](#).

Cas12a and CBE sgRNA design

For optimizing the CBE targeting of tdTomato and PF16, we selected sgRNAs from Engstler and Beneke (2023) [\[1\]](#) (Supplementary File 1).

Cas12a crRNAs were designed using CCTop (Labuhn et al., 2018 [\[3\]](#); Stemmer et al., 2015 [\[4\]](#)) with the “PAM type” setting “TTTN” and species specification “Protists – Euglenozoa – *Leishmania tarentolae*”. For the Cas12a crRNA sequence search, we selected the 18S rRNA locus from *L. mexicana* (using an ONT-Illumina sequencing optimised MHOMGT2001U1103 genome annotation from Beneke et al. (2022) [\[5\]](#)) and selected homology flanks routinely used for integration into this locus (Misslitz et al., 2000 [\[6\]](#); Sorensen et al., 2003 [\[7\]](#)). For designing Cas12a crRNAs that would target a potential “landing-pad”, we searched the neomycin-resistance marker for possible spacer sequences. All resulting Cas12a crRNAs were then checked for possible off-targets by using a local blast search against all *Leishmania* species currently available on TriTrypDB (release 59, (Aslett et al., 2010 [\[8\]](#))). This ensures that Cas12a crRNAs can be used for specific donor DNA integration in these species.

Plasmid construction

All generated plasmids were subjected to whole plasmid sequencing at Plasmidsaurus and critical sites, were additionally subjected to Sanger sequencing.

Primers were all ordered as standard desalted oligos at 25 nmole scale (Sigma). All oligo sequences can be found in Supplementary File 1. Plasmid maps are contained within Supplementary File 2 and can also be downloaded from www.leishbaseedit.net.

Construction of CBE, Cas12a and T7 RNAP co-expression plasmids

We synthesized the open-reading frame (ORF) of AsCas12a ultra (M537R/F870L mutation) (Zhang et al., 2021) with a fused SV40 nuclear localisation signal (NLS), a 3xFLAG tag, a P2A self-cleaving peptide and a c-myc NLS, consisting of six repeats. This c-myc NLS repeat has been previously shown to increase editing activity (Gier et al., 2020). We then amplified the synthesized Cas12a construct in a fusion PCR, adding parts of the T7 RNAP (ORF) and an intergenic region to the construct (using primer-pair 2051F/2052R and 2053F/2054R for amplification from pTB007-hyBE4max). Subsequently, this enabled cloning of the AsCas12a-expression cassette into pTB007-hyBE4max using FseI and AflII. We termed the resulting plasmid pTB107, which allows for the co-expression of AsCas12a ultra, T7 RNAP, hyBE4max CBE and hygromycin resistance-marker. To create a version of pTB107 that contained our previously pioneered *Leishmania*-derived version of the Rad51 single-stranded DNA-binding domain (ssDNA-DBD), we digested pLdCH-hyBE4max-LmajDBD (Engstler and Beneke, 2023) and pTB107 with AvrII and FseI to generate pTB106.

Construction of sgRNA expression plasmids and their optimisation by promoter testing

For generation of T7 RNAP promoter-driven CBE sgRNA expression plasmids, we amplified and cloned the sgRNA-expression cassette from pLdCH into NsiI and MluI sites of a pPLOT-Puro plasmid (Beneke et al., 2017), thereby introducing the unmodified T7 RNAP promoter sequence (amplification using primer-pair 2128F-WT/2129R). To enable sgRNA cloning using BbsI sites, we then eliminated the BbsI site contained within the Actin 5'UTR of this plasmid, using primer-pair 2126F/2129R and PspOMI/MluI restriction sites. To avoid over-expression of the puromycin drug resistance marker, we also eliminated the additional unmodified T7 RNAP promoter sequence contained in the plasmid backbone by amplifying the whole expression cassette (using primer-pair 2067F/2068R) and cloning it using PacI into the bacterial backbone of plasmid pTB007 (Beneke et al., 2017). We then cloned into this resulting plasmid T7 RNAP and rRNAP promoter variant cassettes, containing tdTomato targeting sgRNAs, by using again NsiI and MluI sites (primer-pairs 2014-2021 and 2025-2027). Tdtomato-targeting sgRNA expression constructs were then transfected as episomes into *L. major* parasites that expressed a tdTomato reporter, T7 RNAP and CBE. Following transfection recovery, the proportion of tdTomato-expressing cells was determined by FACS analysis and doubling times measured. Thereby, transfection and FACS analysis were carried out as described previously (Engstler and Beneke, 2023). Following promoter testing, we then finalized our design and generated pTB102 with a T7 RNAP T10 GG promoter variant (using primer-pair 2128F-T10GG/2129R).

To include homology flanks in the pTB102 plasmid, neomycin-targeting or 18S rRNA-targeting regions were amplified and fused to pTB102 sgRNA expression cassettes using a fusion PCR approach (primer-pairs 2236F-2257R). Resulting amplicons were cloned into PacI sites, resulting in pTB104 (neomycin-targeting T7 RNAP expression plasmid) and pTB105 (18S rRNA-targeting T7 RNAP expression plasmid).

PF16 and tdTomato targeting CBE sgRNAs were cloned into pTB102, pTB104 and pTB105 plasmids using BbsI sites as previously described (Engstler and Beneke, 2023).

Preparation of donor DNA and Cas12a crRNA template DNA

To amplify the CBE sgRNA expression donor construct, 50 ng of plasmid template (pTB102, pTB104 or pTB105) containing a CBE targeting guide, 200 μ M dNTPs, 0.5 μ M each of forward and reverse primers, and 1 unit of Q5 polymerase (New England Biolabs) were mixed in 1x Q5 buffer to a final volume of 100 μ l. Oligos used for amplification contained homology flanks of varying lengths. PCR steps were 30 s at 98°C, followed by 20 cycles of 10 s at 98°C, 10 s at 65°C and 40 s at 72°C, concluding with a final elongation step of 5 min at 72°C. Alternatively, donor DNA was produced by digesting pTB104 and pTB105 plasmids with PacI. To confirm the presence of the expected product, 2 μ l of this reaction was analysed on a 0.8% agarose gel.

For the amplification of the Cas12a crRNA template DNA, we used a common forward primer, containing an unmodified T7 RNAP promoter sequence and an optimised Cas12a direct repeat (DR) from DeWeirdt et al. (2021) [\[1\]](#). For the PCR 200 μ M dNTPs, 2 μ M each of Cas12a forward primer and corresponding reverse primer, and 1 unit of Phusion Polymerase (Thermo Fisher) were mixed in 1x Phusion GC Buffer and 3 % DMSO to a total volume of 50 μ l. PCR steps were 30 s at 98°C followed by 35 cycles of 10 s at 98°C, 10 s at 65°C and 10s at 72° concluding with a final elongation step for 7 min at 72°C. Successful amplification was confirmed by running 2 μ l of the reaction on a 2% agarose gel.

The remainder of the donor and crRNA reaction was combined, EtOH purified, resuspended in 50 μ l ultra-pure water, heat sterilised at 95°C for 5 min and then used for transfection.

Testing Cas12a-mediated integration of mNG constructs

To test the Cas12a-mediated integration efficiency of donor DNA constructs with varying homology flank lengths, we generated a DNA construct containing mNeonGreen (mNG) as a reporter and blasticidin as a resistance marker, as well as ~600 nt homology flanks targeting the 18S rRNA SSU locus (Misslitz et al., 2000 [\[2\]](#); Sorensen et al., 2003 [\[3\]](#)). This construct was assembled by fusion PCR as previously described (Engstler and Beneke, 2023 [\[4\]](#)), using pPLOT-mNG-Blast (Beneke et al., 2017 [\[5\]](#)) as a PCR template. The generated fusion construct was then cloned into a blunted vector backbone, resulting in the plasmid pPLOT-SSU-HDR-mNG-Blast.

We then PCR amplified as described above donor DNA from this plasmid with varying homology flank lengths (~600 nt, 60 nt and 30 nt; Supplementary File 1). Resulting PCR amplicons and the pPLOT-SSU-HDR-mNG-Blast plasmid itself were then purified using ethanol precipitation and normalized to the same number of DNA molecules (3.6×10^{12} molecules per transfection). This normalization was based on 10 μ g of pPLOT-SSU-HDR-mNG-Blast plasmid. In addition, 10 μ g of Cas12a crRNA template DNA was added to donor DNA as indicated (Figure 2 [\[6\]](#)). As described previously (Engstler and Beneke, 2023 [\[4\]](#)), DNA mixtures were then transfected using 5×10^6 cells per transfection and we determined the number of blasticidin resistant clones by immediately diluting transfected populations and plating them on 96 well plates. In addition, 10 days post transfection undiluted blasticidin resistant populations were subjected to FACS analysis. To verify the correct integration of donor DNA, DNA was isolated from these populations as described (Engstler and Beneke, 2023 [\[4\]](#); Rotureau et al., 2005 [\[7\]](#)) and amplified integration sites were submitted for Sanger sequencing at Eurofins Genomics (Figure 2 S1A [\[8\]](#) and D [\[9\]](#)).

High efficiency transfections

Transfections were generally carried out as described in Engstler and Beneke (2023) [\[4\]](#), transfecting $\sim 5 \times 10^6$ cells in 1x Tb-BSF buffer (Schumann Burkard et al., 2011 [\[10\]](#)) using 2 mm cuvettes (BTX) with an Amaxa Nucleofector 2b (Lonza, program X-001).

For high efficiency transfections, we modified the protocol and compared site-by-site the transfection efficiency of the Basic Parasite Nucleofector Kit (Lonza) and Tb-BSF protocol (Schumann Burkard et al., 2011 [\[1\]](#)). 1×10^8 or 2.5×10^8 cells were collected, washed once in 1x Tb-BSF buffer and resuspended in either 100 μ l Lonza transfection reagent or 200 μ l Tb-BSF buffer. 10 μ g donor DNA and 10 μ g Cas12a crRNA template DNA were diluted in either 20 μ l (Lonza transfection) or 50 μ l (Tb-BSF transfection) ultrapure water, heat sterilized and added to each transfection respectively (final transfection volume Tb-BSF 250 μ l, Lonza 120 μ l). Cells were transfected using transfection cuvettes supplied with the Basic Parasite Nucleofector Kit (Lonza transfection) or 2 mm BTX cuvettes (Tb-BSF transfection) with one pulse of the X-001 program on an Amaxa Nucleofector 2b (Lonza). For measuring transfection efficiencies, cells were selected and diluted on 96-well plates as described (Engstler and Beneke, 2023 [\[2\]](#)).

Targeting of PF16 with an improved CBE system

For testing the editing efficiency of our improved CBE system, we amplified donor DNA from the pTB102 plasmid, containing the sgRNA PF16-3 from Engstler and Beneke (2023 [\[2\]](#)). As indicated (Figure 4 [\[3\]](#)), donor DNAs were mixed with respective Cas12a crRNA DNA templates and transfected into *Leishmania* parasites. Following selection and the recovery of parasites, mutants were tracked in duplicates or triplicates and their mean velocity determined as described before (Engstler and Beneke, 2023 [\[2\]](#)). Editing rates were measured by amplifying the PF16 target locus and by subjecting PCR amplicons to Sanger sequencing (Engstler and Beneke, 2023 [\[2\]](#)).

ONT and Illumina sequencing analysis of CBE PF16 mutants

DNA from *L. mexicana* CBE PF16 mutants was isolated as previously described (Engstler and Beneke, 2023 [\[2\]](#); Rotureau et al., 2005 [\[4\]](#)) and submitted for ONT and Illumina sequencing at Plasmidsaurus (service: “Big Hybrid ONT + Illumina”). Obtained raw fastq files from ONT and Illumina sequencing were then assembled using the Burrows-Wheeler Aligner (Li and Durbin, 2009 [\[5\]](#)) using two customized *L. mexicana* genomes of the MHOMGT2001U1103 annotation. In the first genome, we duplicated chromosome 27 and modified the 18S rRNA SSU locus of one chromosome 27 copy to include the tdTomato expression cassette (Engstler and Beneke, 2023 [\[2\]](#)) (Figure 4 [\[3\]](#) S1B [\[6\]](#)), while the other copy contained the CBE sgRNA expression construct. In the second genome, we instead modified chromosome 27 to contain both expression cassettes consecutively (Figure 4 [\[3\]](#) S1C [\[6\]](#)). Following alignments, we then used samtools (Li et al., 2009 [\[7\]](#)) for sorting and indexing of bam files and viewed our analysis using the IGV genome browser (Robinson et al., 2011 [\[8\]](#)).

Optimised CBE sgRNA scoring

To determine the likelihood of successful gene editing, we adjusted the sgRNA ranking system. New sgRNA ranking scores were calculated based on various parameters, including the relative likelihood of generating a STOP codon from a cytosine base, the total number of possible STOP codons, the location of cytosines within the editing window, and the position of the guide’s target within the coding sequence (CDS) of the gene. To generate the overall score, we weighted the contribution of each of these scores, with the likelihood of generating a stop codon having four times more contribution than the other scores. We then normalized the total score on a scale of 0 to 100, where 0 represents the best score, and 100 represents the worst score.

The updated data sets and code can be accessed at <http://www.leishbaseedit.net/> [\[9\]](#) and https://github.com/ElisabethMeiser/Collaboration_Beneke_Meiser [\[10\]](#), respectively.

Cas12a-mediated gene replacement for generation of knockout cell lines

For the Cas12a-mediated replacement of the PF16 CDS, we adapted the LeishGEdit approach (Beneke and Gluenz, 2019 [↗](#); Beneke and Gluenz, 2020 [↗](#); Beneke et al., 2017 [↗](#)) and PCR-amplified two Cas12a crRNAs targeting the 5' and 3' UTR of the PF16 ORF (crRNA design as described above). In addition, we PCR-amplified pTBlast and pTPuro cassettes from the LeishGEdit toolbox with primers containing 30 nt homology flanks adjacent to these crRNAs. Both PCRs were carried out as described above and we then co-transfected all resulting products into *L. mexicana* cells containing the pTB107 plasmid. Successful gene replacement was confirmed by amplifying and Sanger sequencing the PF16 locus with two ORF spanning primers.

Availability

All data generated or analysed during this study are included in the manuscript and supplementary file. LeishBASEedit is an open-source primer design tool available under www.leishbaseedit.net [↗](#).

Acknowledgements

We thank all members of the Alsheimer, Engstler, Janzen and Kramer group for helpful discussions and support. In addition, we thank Markus Engstler for providing research resources and helpful comments on the manuscript. This project was funded by Tom Beneke through an EMBO Postdoctoral Fellowship (ALTF 727-2021) and Marie Skłodowska-Curie Actions Postdoctoral Fellowship (101064428 – LeishMOM).

Conflict of interest

All authors declare that they have no conflicts of interest.

Note

This reviewed preprint has been updated to move Figure 5 to the correct point in the text.

Figure legends

Supplementary files

Supplementary File 1. Primers used in this study.

Supplementary File 2. Plasmid maps. Genbank files of pTB102, pTB104, pTB105, pTB106 and pTB107.

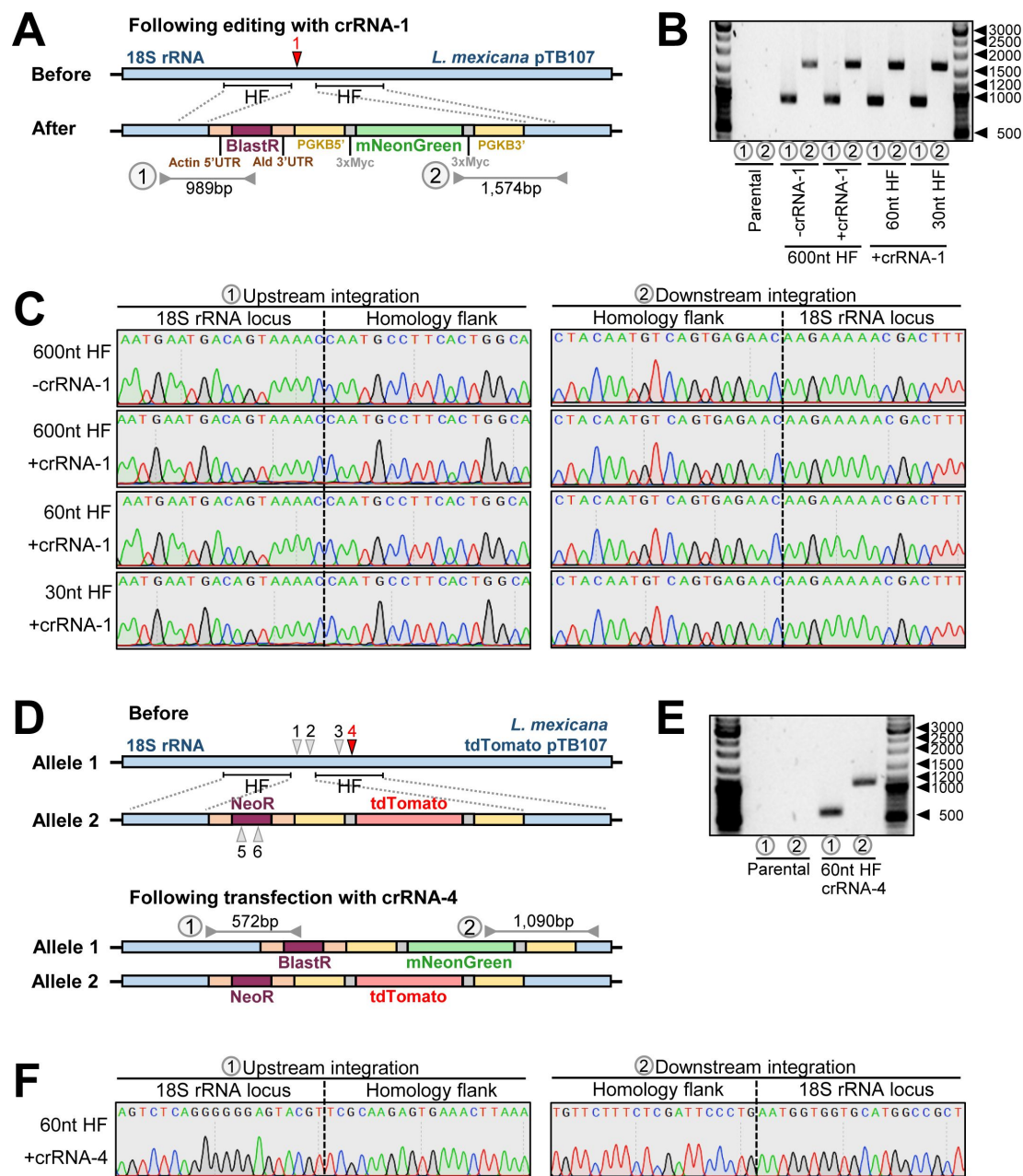


Figure 2 S1.

Verification of AsCas12a-mediated reporter construct integration.

(A) Locus map of the 18S rRNA SSU locus in *L. mexicana* pTB107 cells before and after co-transfection of Cas12a crRNA-1 and a reporter construct (as described in Figure 2A, C and D). The positions of Cas12a crRNA-1 and HF used for the integration of the reporter are highlighted. Amplicons of PCR reactions spanning the integration sites are indicated and amplification results are shown in (B), with DNA sizes highlighted in bp. (C) Sanger sequencing trace plots of amplicons shown in (B). The site of integration into the 18S rRNA locus is highlighted with a dotted line. (D) Locus map of the 18S rRNA SSU locus in *L. mexicana* pTB107 cells expressing tdTomato before and after co-transfection of various Cas12a crRNAs and a reporter construct (as described in Figure 2A, E and F). The positions of Cas12a crRNA-1, 2, 3, 4, 5 and 6 are highlighted. Indicated HF positions have been used for the integration of the tdTomato expression construct on one allele but individual HFs adjacent to crRNA-1 to 6 have been used for integration of the reporter construct. Amplicons of PCR reactions spanning the integration sites are indicated and amplification results are shown in (E) for verification of crRNA-4. (F) Sanger sequencing trace plots of amplicons shown in (E).

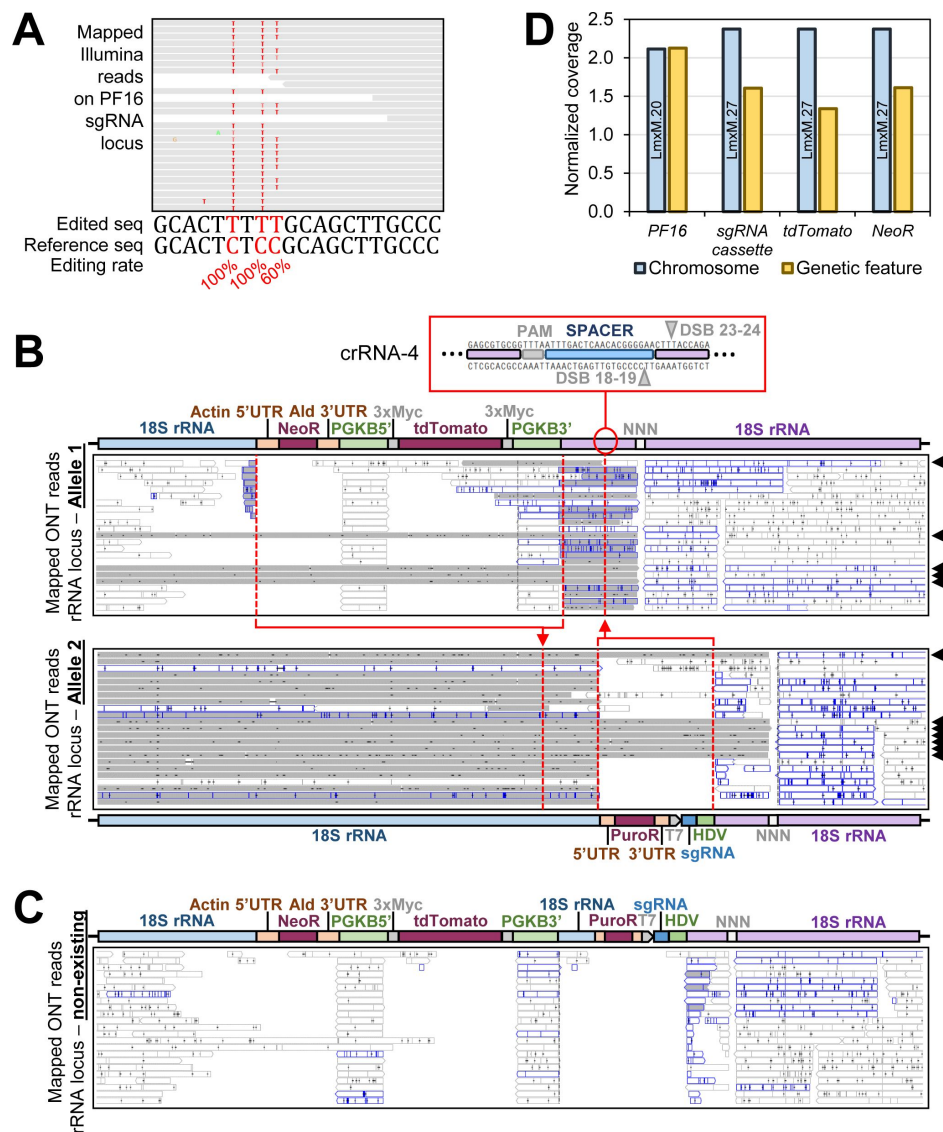


Figure 4 S1.

ONT and Illumina sequencing of transfectants.

L. mexicana pTB107 cells expressing tdTomato were co-transfected with Cas12a crRNA-4 template DNA constructs and a CBE sgRNA expression cassette containing a PF16 targeting guide sequence. Isolated DNA from parasites were subjected to ONT and Illumina sequencing 16 days post transfection. **(A)** Illumina sequencing reads were mapped against the *L. mexicana* MHOMGT2001U1103 genome annotation. The edited consensus sequence is shown alongside with the reference sequence and calculated editing rates (based on 35× genome coverage). **(B and C)** ONT reads were mapped against two customized *L. mexicana* MHOMGT2001U1103 genome annotations and viewed in the IGV genome browser (Robinson et al., 2011), with uniquely mapped reads in grey and reads with multiple alignments in white. Reads mapped against the PGKB 5' and 3'UTR are over-represented as these UTRs are also present in the pTB107 plasmid. **(B)** ONT reads were mapped to a customized genome in which we assumed that the 18S rRNA SSU locus had one integrated copy of the tdTomato expression cassette on one allele and one integrated copy of the CBE sgRNA expression construct on the other allele. Schematics show the annotation of genetic elements assumed to be on each allele. The position of one allele in respect to the position on the other allele is indicated with red arrows. The position of the Cas12a crRNA-4 to integrate the CBE sgRNA expression cassette is additionally highlighted. Black arrows indicate unique ONT reads that cover the entire customized locus. Reads are trimmed as the genome annotation is incomplete (NNN gap highlighted). **(C)** ONT reads were mapped to a customized genome in which we assumed that both cassettes were integrated on the same allele and could be detected on a single ONT read. **(D)** Illumina read coverage was normalized and chromosome coverage was compared to coverage of individual genetic features on these chromosomes.

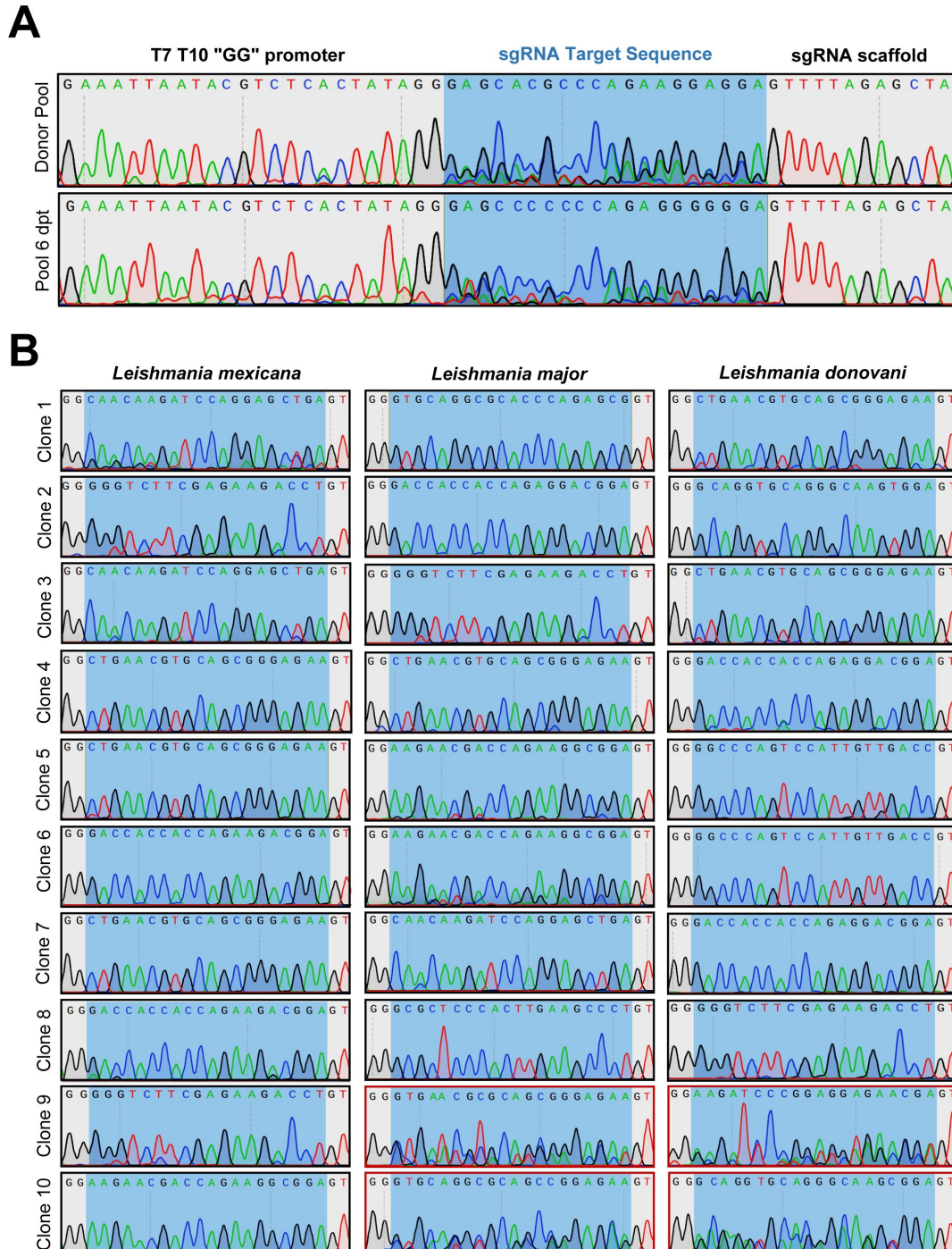


Figure 4 S2.

Sequencing of clones isolated from an AsCas12a-mediated library transfection.

L. mexicana, *L. major* and *L. donovani* tdTomato/pTB107 cell lines were co-transfected with a library containing a mixtures of 15 guides and Cas12a crRNA-4 targeting the 18S rRNA SSU locus. Library clones from library transfection were isolated and their CBE sgRNA amplified for Sanger sequencing.

(A) Sanger sequencing trace plots of the library before and 6 days post transfection. The T7 T10 GG promoter, sgRNA target sequence and the beginning of the sgRNA scaffold sequence are highlighted.

(B) Sanger sequencing trace plots of guide cassettes sequenced from clones. Highlighted in red are clones that maintain more than one sgRNA.

Score		C to STOP ratio
high	high mid low NNNNNNCCNNNNNNNNNNNNNNNNNNNNNN 2xSTOP	1:1
↑	NNNNNCNNNNNNNNNNNNNNNNNNNNNN 1xSTOP	1:1
	NNNNCCNNNCNNNNNNNNNNNNNNNNNNNN 1xSTOP 1xNO-STOP	1:2
	NNNNNCNNNCNCNNNNNNNNNNNNNNNNNN 1xSTOP 2xNO-STOP	1:3
low		

Figure 4 S3.

Schematic of new CBE scoring matrix. A sgRNA receives the highest score, when the ratio of available cytosines to resulting STOP codons is 1:1. Additionally, the final score depends on the number of possible STOP codons and position of cytosines within the editing window (activities indicated above: high, mid and low).

Supplementary File 3. A protocol for AsCas12a-mediated transfection of CBE sgRNA expression cassettes. Step-by-step protocol includes guidelines for PCR amplification of donor DNA and Cas12a crRNA template DNA, as well as preparations for transfection.

Source code 1. Novel scoring for CBE sgRNA sorting.

Source code for CBE sgRNA scoring described above. The updated data sets can be accessed at <http://www.leishbaseedit.net/>. The source code for the guide scoring has been uploaded to https://github.com/ElisabethMeiser/Collaboration_Beneke_Meiser.

Source data 1. Raw DNA images of Figure 2 S1B and E. Details as described in **Figure 2 S1**.

Source data 2. Raw DNA images of Figure 5B. Details as described in **Figure 5**.

References

- Aslett M. *et al.* (2010) **TriTrypDB: a functional genomic resource for the Trypanosomatidae** *Nucleic Acids Res* **38**:D457–62
- Baker N. *et al.* (2021) **Systematic functional analysis of Leishmania protein kinases identifies regulators of differentiation or survival** *Nat Commun* **12**
- Beneke T. *et al.* (2019) **Genetic dissection of a Leishmania flagellar proteome demonstrates requirement for directional motility in sand fly infections** *PLoS Pathog* **15**
- Beneke T., Dobramysl U., Catta-Preta C. M. C., Mottram J. C., Gluenz E., Wheeler R. J. (2022) **Genome sequence of Leishmania mexicana MNYC/BZ/62/M379 expressing Cas9 and T7 RNA polymerase** *Wellcome Open Res* **7**
- Beneke T., Gluenz E. (2019) **LeishGEdit: A Method for Rapid Gene Knockout and Tagging Using CRISPR-Cas9** *Methods Mol Biol* **1971**:189–210
- Beneke T., Gluenz E. (2020) **Bar-seq strategies for the LeishGEdit toolbox** *Mol Biochem Parasitol* **239**
- Beneke T., Madden R., Makin L., Valli J., Sunter J., Gluenz E. (2017) **A CRISPR Cas9 high-throughput genome editing toolkit for kinetoplastids** *R Soc Open Sci* **4**
- Burge R. J., Damianou A., Wilkinson A. J., Rodenko B., Mottram J. C. (2020) **Leishmania differentiation requires ubiquitin conjugation mediated by a UBC2-UEV1 E2 complex** *PLoS Pathog* **16**
- Burkard G., Fragoso C. M., Roditi I. (2007) **Highly efficient stable transformation of bloodstream forms of Trypanosoma brucei** *Mol Biochem Parasitol* **153**:220–3
- Damianou A., Burge R. J., Catta-Preta C. M. C., Geoghegan V., Nievas Y. R., Newling K., Brown E., Burchmore R., Rodenko B., Mottram J. C. (2020) **Essential roles for deubiquitination in Leishmania life cycle progression** *PLoS Pathog* **16**

- de Paiva R. M. *et al.* (2015) **Amastin Knockdown in *Leishmania braziliensis* Affects Parasite-Macrophage Interaction and Results in Impaired Viability of Intracellular Amastigotes** *PLoS Pathog* **11**
- Decuyper S., Rijal S., Yardley V., De Doncker S., Laurent T., Khanal B., Chappuis F., Dujardin J. C. (2005) **Gene expression analysis of the mechanism of natural Sb(V) resistance in *Leishmania donovani* isolates from Nepal** *Antimicrob Agents Chemother* **49**:4616–21
- DeWeirdt P. C. *et al.* (2021) **Optimization of AsCas12a for combinatorial genetic screens in human cells** *Nat Biotechnol* **39**:94–104
- Engstler M., Beneke T. (2023) **Gene editing and scalable functional genomic screening in *Leishmania* species using the CRISPR/Cas9 cytosine base editor toolbox LeishBASEedit** *Elife* **12**
- Gier R. A., Budinich K. A., Evitt N. H., Cao Z., Freilich E. S., Chen Q., Qi J., Lan Y., Kohli R. M., Shi J. (2020) **High-performance CRISPR-Cas12a genome editing for combinatorial genetic screening** *Nat Commun* **11**
- Glover L., Alsford S., Baker N., Turner D. J., Sanchez-Flores A., Hutchinson S., Hertz-Fowler C., Berriman M., Horn D. (2015) **Genome-scale RNAi screens for high-throughput phenotyping in bloodstream-form African trypanosomes** *Nat Protoc* **10**:106–33
- Glover L., Horn D. (2009) **Site-specific DNA double-strand breaks greatly increase stable transformation efficiency in *Trypanosoma brucei*** *Mol Biochem Parasitol* **166**:194–7
- Horn D. (2022) **Genome-scale RNAi screens in African trypanosomes** *Trends Parasitol* **38**:160–173
- Labuhn M., Adams F. F., Ng M., Knoess S., Schambach A., Charpentier E. M., Schwarzer A., Mateo J. L., Klusmann J. H., Heckl D. (2018) **Refined sgRNA efficacy prediction improves large- and small-scale CRISPR-Cas9 applications** *Nucleic Acids Res* **46**:1375–1385
- Li H., Durbin R. (2009) **Fast and accurate short read alignment with Burrows-Wheeler transform** *Bioinformatics* **25**:1754–60
- Li H., Handsaker B., Wysoker A., Fennell T., Ruan J., Homer N., Marth G., Abecasis G., Durbin R., Genome Project Data Processing, S (2009) **The Sequence Alignment/Map format and SAMtools** *Bioinformatics* **25**:2078–9
- Lye L. F., Owens K., Shi H., Murta S. M., Vieira A. C., Turco S. J., Tschudi C., Ullu E., Beverley S. M. (2010) **Retention and loss of RNA interference pathways in trypanosomatid protozoans** *PLoS Pathog* **6**
- Lye L. F., Owens K. L., Jang S., Marcus J. E., Brettmann E. A., Beverley S. M. (2022) **An RNA Interference (RNAi) Toolkit and Its Utility for Functional Genetic Analysis of *Leishmania* (Viannia)** *Genes (Basel)* **14**
- Martel D., Beneke T., Gluenz E., Spath G. F., Rachidi N. (2017) **Characterisation of Casein Kinase 1.1 in *Leishmania donovani* Using the CRISPR Cas9 Toolkit** *Biomed Res Int* **2017**
- Misslitz A., Mottram J. C., Overath P., Aebischer T. (2000) **Targeted integration into a rRNA locus results in uniform and high level expression of transgenes in *Leishmania* amastigotes** *Mol Biochem Parasitol* **107**:251–61

- Morris J. C., Wang Z., Drew M. E., Englund P. T. (2002) **Glycolysis modulates trypanosome glycoprotein expression as revealed by an RNAi library** *EMBO J* **21**:4429–38
- Padmanabhan R., Sarcar S. N., Miller D. L. (2020) **Promoter Length Affects the Initiation of T7 RNA Polymerase In Vitro: New Insights into Promoter/Polymerase Co-evolution** *J Mol Evol* **88**:179–193
- Passos-Silva D. G., Rajao M. A., Nascimento de Aguiar P. H., Vieira-da-Rocha J. P., Machado C. R., Furtado C. (2010) **Overview of DNA Repair in Trypanosoma cruzi, Trypanosoma brucei, and Leishmania major** *J Nucleic Acids* **2010**
- Paul B., Montoya G. (2020) **CRISPR-Cas12a: Functional overview and applications** *Biomed J* **43**:8–17
- Robinson J. T., Thorvaldsdottir H., Winckler W., Guttman M., Lander E. S., Getz G., Mesirov J. P. (2011) **Integrative genomics viewer** *Nat Biotechnol* **29**:24–6
- Rotureau B., Gego A., Carme B. (2005) **Trypanosomatid protozoa: a simplified DNA isolation procedure** *Exp Parasitol* **111**:207–9
- Sanjana N. E., Shalem O., Zhang F. (2014) **Improved vectors and genome-wide libraries for CRISPR screening** *Nat Methods* **11**:783–784
- Schumann Burkard G., Jutzi P., and Roditi (2011) **Genome-wide RNAi screens in bloodstream form trypanosomes identify drug transporters** *Mol Biochem Parasitol* **175**:91–4
- Schumann G., Kangussu-Marcolino M. M., Doiron N., Kaser S., de Assis Burle-Caldas G., DaRocha W. D., Teixeira S. M., Roditi I. (2017) **Zinc finger nuclease technology: A stable tool for high efficiency transformation in bloodstream form T. brucei** *Mol Biochem Parasitol* **213**:12–15
- Sorensen M., Lippuner C., Kaiser T., Misslitz A., Aebischer T., Bumann D. (2003) **Rapidly maturing red fluorescent protein variants with strongly enhanced brightness in bacteria** *FEBS Lett* **552**:110–4
- Stemmer M., Thumberger T., Del Sol Keyer M., Wittbrodt J., Mateo J. L. (2015) **CCTop: An Intuitive, Flexible and Reliable CRISPR/Cas9 Target Prediction Tool** *PLoS One* **10**
- Ullu E., Tschudi C., Chakraborty T. (2004) **RNA interference in protozoan parasites** *Cell Microbiol* **6**:509–19
- Wirtz E., Hoek M., Cross G. A. (1998) **Regulated processive transcription of chromatin by T7 RNA polymerase in Trypanosoma brucei** *Nucleic Acids Res* **26**:4626–34
- Yau E. H., Rana T. M. (2018) **Next-Generation Sequencing of Genome-Wide CRISPR Screens** *Methods Mol Biol* **1712**:203–216
- Zhang L. et al. (2021) **AsCas12a ultra nuclease facilitates the rapid generation of therapeutic cell medicines** *Nat Commun* **12**
- Zhang W. W., Lypaczewski P., Matlashewski G. (2017) **Optimized CRISPR-Cas9 Genome Editing for Leishmania and Its Use To Target a Multigene Family, Induce Chromosomal Translocation, and Study DNA Break Repair Mechanisms** *mSphere* **2**

Zhang W. W., Matlashewski G. (2015) **CRISPR-Cas9-Mediated Genome Editing in *Leishmania donovani*** *MBio* 6

Zhang W. W., Matlashewski G. (2019) **Single-Strand Annealing Plays a Major Role in Double-Strand DNA Break Repair following CRISPR-Cas9 Cleavage in *Leishmania*** *mSphere* 4

Zhang W. W., Wright D. G., Harrison L., Matlashewski G. (2022) **Reconstitution of *Mycobacterium marinum* Nonhomologous DNA End Joining Pathway in *Leishmania*** *mSphere* 7

Zhang X. *et al.* (2020) **Increasing the efficiency and targeting range of cytidine base editors through fusion of a single-stranded DNA-binding protein domain** *Nat Cell Biol* 22:740–750

Editors

Reviewing Editor

Christine Clayton

Centre for Molecular Biology of Heidelberg University (ZMBH), retired, Heidelberg, Germany

Senior Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Reviewer #1 (Public Review):

While CRISPR/Cas technology has greatly facilitated the ability to perform precise genome edits in *Leishmania* spp., the lack of a non-homologous DNA end-joining (NHEJ) pathway in *Leishmania* has prevented researchers from performing large-scale Cas-based perturbation screens. With the introduction of base editing technology to the *Leishmania* field, the Beneke lab has begun to address this challenge (Engstler and Beneke, 2023).

In this study, the authors build on their previously published protocols and develop a strategy that:

- (1) allows for very high editing efficiency. The cell editing frequency of 1 edit per 70 cells reported in this study represents a 400-fold improvement over the previously published protocol,
- (2) reduces the negative effects of high sgRNA levels on parasite growth by using a weaker T7 promoter to drive sgRNA transcription.

The combination of these two improvements should open the door to exciting large-scale screens and thus be of great interest to researchers working with *Leishmania* and beyond.

<https://doi.org/10.7554/eLife.97437.1.sa2>

Reviewer #2 (Public Review):

Summary:

Previously, the authors published a *Leishmania* cytosine base editor (CBE) genetic tool that enables the generation of functionally null mutants. This works by utilising a CAS9-cytidine deaminase variant that is targeted to a genetic locus by a small guide RNA (sgRNA) and causes cytosine to thymine conversion. This has the potential to generate a premature stop codon and therefore a loss of function mutant.

CBE has advantages over existing CAS-based knockout tools because it allows the targeting of multicopy gene families and, potentially, the easier generation of pooled loss of function mutants in complex population experiments. Although successful, the first generation of this genetic tool had several limitations that may have prevented its wider adoption, especially in complex genome-wide screens. These include nonspecific toxicity of the sgRNAs, low transfection efficiencies, low editing efficiencies, a proportion of transfectants that express multiple different sgRNAs, and insufficient effectivity in some *Leishmania* species.

Here, the authors set out to systematically solve each of these limitations. By trialling different transfection conditions and different CAS12a cut sites to promote sgRNA expression cassette integration, they increase the transfection efficiency 400-fold and ensure that only a single sgRNA expression cassette integrates that edits with high efficiencies. By trialling different T7 promoters, they significantly reduce the non-specific toxicity of sgRNA expression whilst retaining high editing efficiencies in several *Leishmania* species (*Leishmania major*, *L. mexicana* and *L. donovani*). By improving the sgRNA design, the authors predict that null mutants will be more efficiently produced after editing.

This tool will find adoption for producing null mutants of single-copy genes, multicopy gene families, and potentially genome-wide mutational analyses.

Strengths:

This is an impressive and thorough study that significantly improves the previous iteration of the CBE. The approach is careful and systematic and reflects the authors' excellent experience developing CRISPR tools. The quality of data and analysis is high and data are clearly presented.

Weaknesses:

Figure 4 shows that editing of PF16 is 'reversed' between day 6 and day 16 in *L. mexicana* WTPB107 cells. The authors reasonably conclude that in drug-selected cells there is a mixed population of edited and non-edited cells, possibly due to mis-integration of the sgRNA expression construct, and non-edited cells outcompete edited cells due to a growth defect in PF16 loss of function mutants. However, this suggests that the CBE tool will not work well for producing mutants with strong fitness phenotypes without incorporating a limiting dilution cloning step (at least in *L. mexicana* and quite possibly other *Leishmania* species). Furthermore, it suggests it will not be possible to incorporate genes associated with a growth defect into a pooled drop-out screen as described in the paper. This issue is not well explored in the paper and the authors have not validated their tool on a gene associated with a severe growth defect, or shown that their tool works in a mixed population setting.

Although welcome, the improvements to the crRNA CBE design tool are hypothetical and untested.

The Sanger and Oxford Nanopore Technology analyses on integration sites of the sgRNA expression cassette integration will not detect the mis-integration of the sgRNA expression construct into an entirely different locus.

<https://doi.org/10.7554/eLife.97437.1.sa1>

Reviewer #3 (Public Review):

Genetic manipulation of *Leishmania* has some challenges, including some limitations in the DNA repair strategies that are present in the organism and the absence of RNA interference in many species. The senior author has contributed significantly to expanding the available

routes towards *Leishmania* genetic manipulation by developing and adapting CRISPR-Cas9 tools to allow gene manipulation via DNA double-strand break repair and, more recently, base modification. This work seeks to improve on some limitations in the tools previously described for the latter approach of base modification leading to base change.

The work in the paper is meticulously described, with solid evidence for most of the improvements that are claimed: Figure 1 clearly describes reduced impairment in the growth of parasites expressing sgRNAs via changes in promoters; Figures 2 and 3 compellingly document the usefulness of using AsCas12a for integration after transformation; and Figures 1 and 4 demonstrate the capacity of the combined modifications to efficiently edit a gene in three different *Leishmania* species. There is little doubt these new tools will be adopted by the *Leishmania* community, adding to the growing arsenal of approaches for genetic manipulation.

There are two weaknesses the authors may wish to address, one smaller and one larger.

(1) The main advance claimed here is in this section title: 'Integration of CBE sgRNA expression cassettes via AsCas12a ultra-introduced DSBs increase editing rates', with the evidence for this presented in Figure 4. It is hard work in the submission to discern what direct evidence there is for editing rates being improved relative to earlier, Cas9-based approaches. Did they directly compare the editing by the new and old approach? If not, can they more clearly explain how they are able to make this claim, either by adding text or a new figure? A side-by-side comparison would emphasise the advance of the new approach more clearly.

(2) The ultimate, stated goal of this work is (abstract) to 'enable a variety of loss-of-function screens', as the older approach had some limitations. This goal is not tested for the new tools that have been developed here; the experiment in Figure 5 merely shows that they can, not unexpectedly, make a gene mutant, which was already possible with available tools. Thus, to what extent is this paper describing a step forward? Why have the authors not run an experiment - even the same one that was described previously in Engstler and Beneke (2023) - to show that the new approach improves on previous tools in such a screen, either in scale or accuracy?

<https://doi.org/10.7554/eLife.97437.1.sa0>

Author response:

We would like to thank all reviewers and editors for their thorough peer review and valuable suggestions. In these provisional responses, we summarize the main concerns raised by the reviewers and outline our planned revisions to address them in the manuscript.

Overall, we are pleased to note that the reviewers agree on the potential value of our updated toolbox for gene editing, highlighting its various applications. However, they also raised several valid concerns, which we have summarized and responded to as follows:

(1) Mutant phenotypes in transfected populations can be occasionally reversed or escaped. This suggests it will not be possible to detect growth-associated phenotypes in pooled screens. An experiment with a pooled loss-of-function screen to test this is missing.

Escapes or reversals of mutant phenotypes have been observed with other genetic tools used for loss-of-function screening, including lentiviral CRISPR approaches in mammalian systems and RNAi in *Trypanosoma brucei*. Cells can escape phenotypes through various mechanisms, such as promoter silencing or selection of non-deleterious mutations. Additionally, not every

CRISPR guide is efficient in generating a mutant phenotype, and RNAi constructs can also vary in their effectiveness. Despite these challenges, genome-wide loss-of-function screens have been successfully carried out in mammalian cells and *Trypanosoma* parasites. Therefore, we believe that the observed escape of one mutant phenotype does not preclude the detection of growth-associated or other phenotypes in pooled screens. Moreover, we did not observe a reversal of the mutant phenotype in *L. mexicana*, *L. donovani*, and *L. major* parasites expressing tdTomato from an expression cassette integrated into the 18S rRNA SSU locus (Figure 4). However, the reviewers are rightfully requesting a pooled loss-of-function screen to validate this. Since submitting this manuscript, we have conducted multiple pooled loss-of-function screens, which have confirmed the ability of our here presented method to detect a range of mutant phenotypes in pooled screening formats. We will include these results in our revised manuscript.

| *(2) The possibility of mis-integration of the CBE sgRNA expression construct into an entirely different locus is not explored.*

We plan to reanalyze our ONT sequencing data to verify if the CBE sgRNA expression construct was integrated into an unintended loci. If we detect any mis-integration events, we will evaluate their potential negative impacts and discuss these findings in the revised manuscript.

| *(3) The achieved increase in editing efficiency compared to the previous base editing method could be more clearly presented.*

We have directly compared our improved method to our previous base editing method in Figures 1E and 4, demonstrating higher editing rates in a much shorter time. In the revised manuscript, we will present and describe the increase in editing rate more clearly.

| *(4) The improvements on CBE sgRNA guide design are hypothetical and untested.*

We agree that the improvements to the CBE sgRNA design are currently hypothetical. We plan to systematically test our guide design principles in future studies. Since this will require testing hundreds of guides to draw robust conclusions, we believe that this aspect is beyond the scope of the current study. However, we will discuss our plans for future validation in the revised manuscript.

Overall, we appreciate the reviewers' insights and are committed to addressing their concerns thoroughly. We believe that the planned revisions and additional experiments will significantly strengthen our manuscript and provide a more comprehensive evaluation of our updated gene editing toolbox.

<https://doi.org/10.7554/eLife.97437.1.sa4>